Employment Protection and Consumption: Evidence from Italy

Nello Esposito*
University of Naples Federico II and the Central Bank of Ireland

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Abstract

Leveraging an Italian labor market reform known as the *Jobs Act* (JA), I study how employment risk affects household consumption and labor supply. The JA reduced protection against individual termination only for workers hired on or after March 7, 2015, by firms with at least 15 employees. Using this time-based discontinuity in employment protection as exogenous variation in employment risk, I find that workers subject to the reform consume 8% less than workers hired through March 6, 2015, who were not exposed to the reform. By contrast, I find no significant effect on weekly hours worked. I use this empirical result to estimate a standard incomplete-markets model with ex-ante heterogeneity in employment risk. Replicating the empirical analysis in a simulated dataset and controlling for permanent income, I find that risk explains about one third of the reduced-form effect on consumption. Finally, I use the model to show that providing more generous unemployment insurance to workers exposed to the reform and less to unexposed workers, keeping the government's budget constraint fixed, maximizes welfare and reduces consumption inequality.

Keywords: Precautionary Saving, Employment Risk, Simulated Method of Moments
JEL Classification: D14, D15, E21, J28, G51

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1 Introduction

People deeply value job security (Baghai et al. 2023). Rothwell and Crabtree (2019) surveyed a representative sample of U.S. employees and asked them to rate the importance of job security for their quality of life on a scale from 1 to 5. Over 90% assigned a score of 4 or 5, indicating strong preferences for stable employment. More recently, BCG (2023) surveyed 11,000 employees across both advanced and developing countries and found that job stability is prioritized even above pay levels. Focusing on Italy - the Country this paper is based on - CENSIS (2025) shows that 50% of Italian private sector workers would prefer to move to the public sector due to higher job stability (Fuchs-Schündeln and Schündeln 2005).

Following the 2008 Financial Crisis, many advanced economies - especially in Southern Europe - implemented labor market reforms that reduced firing costs for firms, with the ultimate purpose of increasing job mobility, productivity, and employment (OECD 2016). From the workers' perspective, however, these reforms often reduced employment protection and, consequently, increased employment risk.

In this paper, I examine how employment protection affects household consumption and labor supply. To this purpose, I leverage the *Jobs Act* (JA), an Italian labor market reform that applies only to workers hired after March 6, 2015, by firms with at least 15 employees. Workers hired before that date remain under the previous regime, called the *Chart of Labor* (COL). Under the COL, employees unlawfully terminated are reinstated to their previous job at the same wage. In contrast, JA workers are not reinstated in case of unjust termination; instead, they receive a lump-sum monetary payment proportional to their tenure. The reform therefore significantly reduced job protection for the latter by nearly eliminating the probability of reinstatement in cases of unlawful dismissal.

While previous literature has examined the role of idiosyncratic income risk in households' savings (Fuchs-Schündeln and Schündeln 2005, Aiyagari 1994), fewer papers have focused specifically on employment risk; this distinction is particularly important since income fluctuations might be driven both by individual choices and different types of shocks (Guiso et al. 2002). Each shock may be differentially insurable and may have different welfare implications, leading to different policy recommendations (Jappelli and Pistaferri 2017 and Low et al. 2010). My paper's key contribution is to isolate the employment risk channel, using a labor market reform that reduced workers' protection and increased employment risk.

I use data from the 2014 and 2018 waves of the *Participation, Labor, Unemployment Survey* (PLUS), conducted by the Italian Research Agency INAPP (*Istituto Nazionale per le Analisi delle Politiche Pubbliche*). PLUS is a representative survey of the Italian population aged 18 to 75 that provides detailed information on demographic characteristics, food consumption, weekly hours worked, employees' firm size, and whether individuals were hired before or starting from March 7, 2015. This last piece of information lets me identify private-sector workers in large firms whose contracts are regulated by either the *Jobs Act* or the *Chart of Labor*.

My baseline finding is that JA workers reduce their food consumption to the household

income rate (the average propensity of food consumption) by 1.5 percentage points. To estimate the *Jobs Act* 's impact on broader measures of consumption, I propose two strategies. First, I adopt an approach similar to Broda and Parker (2014), in which I scale the effect on the average propensity to food consumption by multiplying it by the ratio of total spending to food spending, as obtained from the Bank of Italy *Survey of Household Income and Wealth* (SHIW). Using this approach, I find an average increase in the overall saving rate of around 4 points. Alternatively, following the approach of Blundell et al. (2008) and Patterson (2023) - BPP from now on - I impute consumption from the Italian *Household Budget Survey* (from now on HBS, the Italian counterpart of the CEX in the United States, containing granular information on household spending) into PLUS, finding a causal effect on this imputed measure of consumption of 8%.

The effect is significant only for people living in the North, Italy's richest and most developed region, compared to the South and the Center, which are relatively underdeveloped. Several factors may explain this geographical heterogeneity. First, informal insurance by parents and relatives might be stronger in the South than in the North, weakening the precautionary saving motive. The second reason might be that people in the South and in the Center are more likely to be hand-to-mouth and therefore off the Euler equation (Christelis et al. 2020). For this reason, prudence plays little role in their saving decisions. While the data do not allow me to investigate the first channel further, I provide evidence supporting the second hypothesis. First, using data from the Bank of Italy *Survey of Household Income and Wealth* (SHIW), I document that the share of hand-to-mouth individuals is higher in the South than in the North, even after accounting for differences in sectoral composition (i.e., sectoral fixed effects). Second, the *Jobs Act* has a stronger effect on consumption for higher-income and more educated households, regardless of where they live.

To strengthen the credibility of the causal claim, I conduct the following placebo tests. First, I show that JA and COL workers' incomes are not statistically different. Second, I run the same analysis as before for workers employed in small firms, where employment protection remained unchanged, and find no significant effects. Third, I examine public sector employees, who are arguably less exposed to employment risk (Fuchs-Schündeln and Schündeln 2005), and for whom employment protection against termination does not apply. Similarly, I find no significant results.

Additionally, I check that hiring rates before and after the reform exhibit a smooth trend, suggesting no manipulation of hiring dates by either workers or firms and no anticipatory effects of the time-based discontinuity in employment protection. Finally, I exclude employees working for firms employing from 15 to 20 employees. The rationale is that the reduction in employment protection was coupled with a hiring subsidy that benefited mostly small firms (Sestito and Viviano 2018 and Boeri and Garibaldi 2019). Marginal firms, such as those with 14 employees, might have decided to hire an additional worker - and thus reach the threshold of 15 employees - to exploit the hiring subsidy. In contrast, firms already well above the threshold faced no such incentives. Results remain economically and statistically significant, indicating that potential

firms' dimensional sorting does not compromise the identification strategy.

The *Jobs Act* impacts consumption via two channels. On the one hand, higher employment risk implies more volatile income and a stronger precautionary saving motive. This is the channel whose importance I aim to measure in this paper. On the other hand, a higher job separation rate reduces the present discounted value of income (PDV), which in turn reduces consumption. In other words, the *Jobs Act* does not induce a mean-preserving spread in individuals' income distributions. Unfortunately, because PDV is unobservable in my data, I cannot empirically disentangle these two mechanisms.

I therefore rely on a variant of the Bewley–Huggett–Aiyagari model (Bewley 1986, Huggett 1993, Aiyagari 1994) - from now on, BHA - in which two types of workers are ex-ante heterogeneous in employment risk: safe and risky workers. The latter mimic JA workers and face a higher separation rate (i.e., the probability of losing their job while employed) than the former, who instead mimic COL workers. I use the difference in consumption between JAs and COLs, together with moments of the cross-sectional wealth distribution and of the labor market, as target moments in the Method of Simulated Moments (MSM) to estimate the risk aversion parameter (assuming preferences with constant relative risk aversion), the discount factor, and the separation rate for both categories of workers. The model matches the empirical target, with estimated preference parameters well within the range reported in previous studies. In my baseline model, the estimated relative risk aversion coefficient is 1.5, while the discount factor is 0.97. I also find that employment risk for JA workers is more than twice as big as the risk faced by COL workers: a separation rate of 4.7% for COLs versus around 12% for the JAs.

I then use the model for two purposes. First, as stated above, I decompose the total effect into the "pure" effect of risk and the variation in permanent income induced by the reform. To this purpose, I simulate a dataset of consumption, income, and wealth. I then regress log consumption on the simulated data on the *Jobs Act* dummy, assets, and the discounted sum of incomes people earn in the model, finding that the effect of employment risk alone is around one third of the JA's total effect on consumption.

I consider the following three extensions of the baseline model. First, I assume recursive preferences à la Epstein and Zin (1989) rather than constant relative risk aversion and estimate the elasticity of intertemporal substitution and risk aversion separately. I find reasonable values for both: the estimated EIS is 0.29, the risk aversion is 1.74, and the discount factor is 0.95. Second, I allow households to save in two account types: liquid and illiquid, with the latter offering higher returns but subject to infrequent adjustment opportunities, as in Auclert et al. (2024). In line with Graves (2025), I find that JA workers - exposed to higher employment risk - increase their savings more in the liquid account than in the illiquid account. The intuition behind this last result is the same as in the seminal paper of Kaplan and Violante (2014): consumers prefer to save, for precautionary reasons, in low-transaction-cost, ready-to-liquidate assets, since they can easily access them when needed.

I also consider a life cycle version of the model in which all individuals start working at 25 years old and under the COL. If they lose their job, the new one they find is regulated by the

JA. Therefore, unlike the baseline model above, in which workers are ex ante set either in the JA or in the COL, in the life cycle model the type of employment is an absorbing state that materializes if the individual loses her job at least once in her working lifetime. This life-cycle structure adds realism to the model, as the *Jobs Act* regulates all *new* employment contracts started on or after March 7, 2015. I again estimate risk aversion and the discount factor via MSM, targeting the empirical effect of interest and the wealth-to-income age profile of workers. Estimates of the parameters are close to the baseline model: risk aversion is 1.48, and the discount factor is 0.99, similarly to Daminato and Padula (2024).

The second purpose is a policy counterfactual. Using the baseline model (with infinite horizon, one asset and CRRA preferences), and following Pellegrini (2025) and Zanella Cavallero et al. (2026), I solve for the type-specific schemes of unemployment insurance (UI) for JAs and COLs that maximize welfare subject to invariant government spending. I find that optimal UI is higher for JA workers than for COL workers, consistent with the intuition that more insurance should be provided to workers more exposed to risk, and less to workers with lower exposure to risk. However, this result comes with one caveat: in the model, employment states are exogenously determined, and not chosen by individuals. This means that there are no moral hazard effects on job search intensity.

This paper carries two policy implications. First, it shows that when designing reforms to address supply-side issues, policymakers must account for their demand-side effects. The Jobs Act aimed to reduce firing costs for firms and foster job creation; it also affected households' exposure to background risk and, consequently, their saving decisions. This highlights that labor market reforms can influence macroeconomic outcomes not only through firms' labor demand, but also through households' consumption and precautionary saving behavior. The second policy message is that job market duality contributed to consumption inequality. The *Jobs Act* introduced an additional layer of duality within open-ended contracts, on top of the existing divide between fixed-term and open-ended contracts (see Daruich et al. 2023 and Genovese 2025).

Related Literature and Contribution. This paper is connected to the literature on income risk and precautionary saving[1]. Empirically assessing the relevance of precautionary saving has proven challenging for two reasons. First, consumption growth volatility, the term governing prudence in the Euler equation (see Fagereng et al. 2017), is unobservable. Therefore, income risk is often used as a proxy, assuming that income is the main source of risk individuals face[2]. In turn, Guiso et al. (2002) argue that income risk depends on the distribution of

[1] Blanchard and Mankiw (1988), Kimball (1990), Caballero (1991), Deaton (1991), Carroll (1997) among the seminal contributions.

[2] See, for instance, Jappelli and Pistaferri (2000). More recently, Guiso and Jappelli (2024a), using the novel *Italian Survey of Consumer Expectations* (ISCE, Guiso and Jappelli 2024b), relate subjective consumption growth risk to earnings, health, energy, and house price risk, as well as aggregate sources of risk such as GDP, interest rate, and inflation risk. Idiosyncratic factors explain about 75% of the variability in consumption risk. In earlier work, Dynan (1993) replaces volatility with realized squared consumption growth and instruments it with lagged demographic characteristics. However, these are predictors of the consumption path, and therefore

future earnings, the distribution of unemployment benefits, and the probability of job loss (i.e., employment risk). While previous studies often bundle employment and income risk together (see Fuchs-Schündeln and Schündeln 2005), this paper distinguishes employment risk (i.e., being under the JA or the COL) from the remaining components of income risk, which mostly depend on idiosyncratic productivity.

The reason it is important to specifically examine employment risk, as opposed to income risk more generally, is the need to separate the effect of exogenous shocks experienced by workers (such as involuntary unemployment) from the endogenous responses to them (such as decisions to stay in or exit the labor market), as the latter carry completely different welfare implications than the former. In this regard, Low et al. (2010), using a life-cycle model with endogenous employment participation and search frictions in the labor market, show that failing to account for job mobility (i.e., the endogenous component of income volatility) overestimates income risk.

The second challenge is that employment risk is endogenous, because workers self-select into riskier and safer jobs according to unobservable characteristics, of which preference heterogeneity is the most prominent example (see Jappelli and Pistaferri 2020).

The literature has followed four approaches to estimate the strength of precautionary saving. The first consists in empirically testing the Euler Equation using survey data to estimate the elasticity of intertemporal substitution (*EIS*) and/or the coefficient of relative prudence (also called the Kimball coefficient from Kimball 1990[3]). Among the papers following this approach are: Guiso and Jappelli (2024a), Bertola et al. (2005) in Italy, and Christelis et al. (2020) in the Netherlands, Sciacchetano (2024) using data on Germany, France, Italy, Spain, the Netherlands and Belgium, and Crump et al. (2022) in the United States. They all find a CRRA coefficient around 1 and 2, corresponding to a prudence parameter between 2 and 3.

The second approach is structural estimation. Gourinchas and Parker (2002) simulate a life cycle model with CRRA preferences and idiosyncratic income risk, and estimate the prudence coefficient (jointly with the discount rate) by minimizing the distance between the average simulated and actual life cycle consumption profiles. Their main result is a risk aversion of 1, which corresponds to a Kimball coefficient of 2. Cagetti (2003), matching median consumption in place of the mean, finds a risk aversion coefficient around 4 (and a Kimball coefficient of 5).

The third approach directly leverages survey questions to elicit motives for saving. Jappelli and Pistaferri (2025) show that households revise their target wealth in response to permanent income shocks at approximately a one-to-one rate. They also show that households converge to wealth faster when they are below target wealth than when they are above. Finally, they simulate a buffer-stock model with different calibration choices and find that reasonable preference and income-process parameters can replicate their results. They elicit target wealth with a direct question in the SHIW, asking how much people think their savings need to be to meet

violate the exclusion restriction.

[3]Assuming CRRA utility, as in the papers cited here, it can be shown that $EIS = 1/\gamma$ and $p = 1 + \gamma$, where γ is the utility function curvature and p is Kimball's coefficient of relative prudence.

unexpected events.

Lusardi (1997) and Guiso et al. (1992) follow a similar strategy, using self-reported measures of income uncertainty to show that subjective income risk is negatively associated with consumption. Graves (2025) finds that higher subjective unemployment risk increases liquid wealth but reduces illiquid wealth. The intuition is that illiquid assets are poorly suited to smoothing income fluctuations, as they are riskier and characterized by transaction costs and/or infrequent adjustment probability (Kaplan and Violante 2014, Auclert et al. 2024). He also shows, using a two-asset HANK model à la Kaplan et al. (2018), that this "flight to liquidity" amplifies business cycle fluctuations.

My paper differs in that it leverages a quasi-experiment rather than subjective expectations of job loss, which might be endogenous to unobservable, potentially time-varying characteristics such as health status and ability. For instance, a worker might expect to be fired soon because her productivity has persistently declined over the last few months and therefore start saving more before unemployment occurs. If so, the increase in savings might be attributed more to the intertemporal motive than to the precautionary saving motive.

The last approach estimates the causal effect on savings of exogenous shocks to income risk or changes in insurance opportunities. Kantor and Fishback (1996) exploit the introduction of workers' compensation in the United States in the 1910s, finding that the introduction of accident insurance decreased households' savings by 25%[4]. Leveraging the historical event of the German reunification in 1990, Fuchs-Schündeln and Schündeln (2005) find that civil servants in Eastern Germany (i.e., former DDR government employees) tend to accumulate 25% less wealth than private workers, controlling for permanent income and demographics. They also demonstrate that not accounting for workers' self-selection underestimates the strength of precautionary saving by half. While the reduced-form approach lets me estimate causally identified effects, it is not very informative about the key structural parameters that underpin them.

To the best of my knowledge, this is the first paper to leverage a time-based discontinuity in employment protection to empirically assess the impact of employment risk on consumption. Beyond focusing on employment risk rather than general income, my paper connects the approaches outlined above from an operational point of view: I estimate the model's structural parameters via MSM, targeting the estimated causal effect of a decrease in employment protection on consumption.

Finally, I contribute to the existing empirical literature by leveraging the *Jobs Act* as a natural experiment. Early work by Boeri and Garibaldi (2019) and Sestito and Viviano (2018) examines the effect of the JA on firms' hiring decisions, finding an increase in the separation rate induced by the reform and an increase in the conversion rate of fixed-term contracts to open-ended contracts. De Paola et al. (2021, 2024) show that the reform reduced women's

[4]There is another strand of the literature that, instead of assuming incomplete markets, tries to microfound them in terms of a principal-agent problem where there is "hidden trade" from the agent side and therefore moral hazard arises. Therefore, the introduction of insurance might be seen as a shock that "alleviates" moral hazard problems, increases insurance opportunities, and reduces the gap with the complete market benchmark.

fertility, both directly (the former paper) and indirectly via workplace peer effects (the latter paper). Bertoni et al. (2023) find that workers hired with the JA perceive their jobs as more precarious and less satisfactory. This prompts workers to engage in on-the-job searching more actively, resulting in higher job mobility rates. Mistrulli et al. (2023) find that being covered by the JA reduced the loan-to-value ratio for mortgages. My incremental contribution to this literature is that efforts to increase labor market flexibility might have second-order effects on savings, in addition to fertility and access to credit.

Structure of the Paper. The paper continues as follows. Section 2 describes the institutional context and the content of the reform. Section 3 describes the PLUS dataset and the sample selection. Section 4 lays out the empirical approach, Section 5 shows the baseline empirical results. Section 6 shows a battery of placebos and robustness checks. It also presents some further results in which I control for risk aversion and some split-sample regressions to investigate relevant heterogeneity dimensions. Section 7 shows the results of the model. Section 8 derives the optimal unemployment insurance policy that maximizes total welfare, subject to the constraint that spending equals the baseline case. Section 9 concludes and outlines directions for further research.

2 Institutional Background and Job Mobility in Italy

This Section explains the reform of the *Jobs Act* and its main consequences for workers. Appendix E provides a detailed description of how Italy regulates employment protection.

The Reform. The backbone of Italy's labor market regulation is the *Chart of Labor* (COL), which applies to all employees of large firms (i.e., with more than 15 employees) hired until March 6, 2015. Under the COL, workers must be reinstated in their jobs in all cases of unjust dismissal, as determined by a labor judge. This entitles them to return to their position, receive compensation for lost wages during their absence, and collect a fine paid by the employer. As a result, this regulation makes dismissals both difficult and costly for employers. Indeed, labor judges in Italy tend to rule in workers' favor in most cases (Del Punta 2025).

The major novelty of the *Jobs Act* (JA), which applies to employees hired by large firms starting from March 7, 2015, is that reinstatement in the case of unjust dismissal is only granted in instances of discriminatory firing - that is, when a worker is dismissed due to gender, race, sexual orientation, or political or religious beliefs. In all other cases of unjust dismissal, as determined by a judge, the employer is not required to reinstate the worker but must pay one-time compensation proportional to the worker's seniority within the firm, subject to a cap.

This change has two main consequences[5]. First, it significantly reduces employment protection in the current job, because it's easier and cheaper for firms to fire workers. To see why, suppose a firm wants to let a COL worker go. It knows that the worker will probably challenge

[5] Table B1 in Appendix B compares the two regimes.

the dismissal in court and that, if the judge rules in the worker's favor, the firm must reinstate the worker. In this scenario, the firm's cost is the compensation the worker would have received in the workplace that the firm no longer wanted to pay (because it wanted to fire the worker), plus a lump-sum payment. By contrast, if the worker was hired starting March 7th, the firm knows that even if the court rules in the worker's favor, it does not have to reinstate the worker, and compensation is capped. As a result, dismissal costs are both lower and more predictable, implying weaker job protection from the worker's perspective. Second, JA workers are more likely to bear job-displacement costs, i.e., long-lasting income losses due to unemployment. This latter point has proven particularly important and has been highlighted by Jarosch (2023) and Caratelli (2024), who show wages are substantially lower after years of unemployment. Moreover, Bertheau et al. (2023), using administrative employee-employer matched data from several advanced economies, compares earnings up to five years after displacement and finds that Italy is among the countries with the most severe and persistent negative effects of job loss.

How did the JA change Employment Protection in Italy? Bertoni et al. (2023) document that individuals hired under the JA are more concerned about losing their jobs compared to those hired under the COL. Moreover, using Italian administrative data, they show that job mobility increased following the introduction of the JA, as workers, especially in low-paying sectors, seek better employment opportunities. In other words, workers tend to change employers more quickly than before, particularly in the early months of employment.

Figure 1 shows that average job tenure (residualized from a quadratic time trend) declined after the implementation of the JA, indicating that in the five years preceding the pandemic, workers' attachment to long-term employment weakened.

Figure 2, on the other hand, presents the OECD Employment Protection Legislation (EPL) index in Italy, France, Germany, and Spain. The EPL index measures how strongly protected workers are in a country. Before the reform, Italy's EPL was the highest among the four largest European Union economies. However, as a consequence of the *Jobs Act*, Italy's EPL score declined sharply and fell below that of France and Germany in 2016. In addition, the 2015/2016 drop is much more pronounced than the 2011/2012 one, due to the Fornero reform.

However, neither the COL nor the JA regulates mass layoffs such as those resulting from firm downsizing or closures. Instead, both laws govern cases in which employers dismiss workers for individual-specific reasons (e.g., disciplinary action or declining individual productivity). Therefore, this analysis does not address the impact of mass layoffs on employment or spending. During the period considered, reforms to the regulation of mass layoffs also applied to all large-firm workers, regardless of their hire date. Therefore, they are not correlated with the *Jobs Act*.

3 Data and Measures of Spending

Dataset Description and Sample Selection. To conduct the analysis, I use INAPP's *Participation, Labour and Unemployment Survey* (PLUS). PLUS is a representative cross-section of the Italian population aged 18 to 75 and contains detailed demographic and labor market information. It is particularly well-suited for my purposes and my context because it explicitly asks respondents whether they were hired before or after March 7, 2015, allowing me to precisely identify JA workers[6]. In terms of outcome variables, the survey contains information on the amount of money spent on at-home food consumption (i.e., groceries) as a share of family income (the average propensity to consume food) and weekly average working hours, i.e., the number of hours they spend at work in a normal week. The 2018 wave of the survey additionally includes questions on risk tolerance (the inverse of risk aversion) and patience, which I use to control for unobserved preference heterogeneity (Jappelli and Pistaferri 2020) in a robustness check. These questions are reported in Appendix A.

Unfortunately, PLUS contains scant information on wealth. There is information on whether households are homeowners or not, which is a proxy for illiquid wealth given that housing represents - together with deposits - the most important form of wealth in Italy, according to Banca d'Italia (2026). However, this information is not particularly useful in this context, as housing is not suitable for insuring against employment shocks, as argued by Kaplan and Violante (2014). However, the question asks how much unforeseen spending households could undertake without relying on external help (i.e., loans from friends, relatives, or parents). The question wording is reported in Appendix A. I interpret this question as a proxy for liquid wealth.

For the main analysis, I retain only currently employed individuals. I then exclude workers whose employer is a public organization (direct civil servants, employees of state-owned companies and utilities, or public independent agencies) and those on fixed-term contracts, as the *Jobs Act* applies exclusively to open-ended contracts. This yields a pooled sample of approximately 8,000 observations. I then use public employees (as well as small firm workers) for the placebo analysis. Table 1 reports descriptive statistics for the full sample and separately for employees in small and large firms. Individuals under 25 make up around 10% of the sample, while those aged 30 to 49 comprise roughly 60%. Males are overrepresented in large firms, and family size differs little. Education levels are also fairly evenly distributed. Both income and weekly working hours are higher in larger firms. Finally, the share of income spent on food is slightly lower in larger firms, likely reflecting their higher average incomes.

Table 2 compares workers in large firms hired under either the COL or JA regime. The most notable difference between the two groups is age: about 30% of JA workers are under 30, compared to just 7% in the COL group. This reflects the fact that younger individuals are more

[6]In principle, a worker's contract could be regulated by the COL even after March 7, 2015, if the employer agrees. This would imply that the estimated causal effect shall be seen as an *intention-to-treat* effect. However, this possibility is, in actuality, very rare. Therefore, being hired after March 7 and being subject to the JA are almost collinear among large firm workers.

likely, by construction, to be hired under the JA regime. Nevertheless, a considerable share of JA workers are over 40. There is no significant gender difference between the two groups, with males comprising roughly two-thirds of each sample. Geographic distribution is also broadly similar. High-education workers make up a substantially larger share of JA workers than COL workers, likely because JA workers are younger and therefore more educated. Average net monthly individual income is nearly identical across the two groups. This matters because it indicates no compensating wage differentials to offset the higher employment risk faced by JA workers. Finally, JA workers report working more hours on average than COL employees.

Scaling Food Consumption. To the best of my knowledge, PLUS is the only available Italian dataset that simultaneously provides information on (i) the legal regime governing unlawful termination (COL or JA), (ii) firm size (above or below 15 employees), and (iii) consumption. However, the consumption variable refers only to monthly at-home food expenditures. Following Broda and Parker (2014), I rescale the effect on food consumption using the average share of food in total consumption, as derived from the Bank of Italy SHIW. This scaling arises from a manipulation of the household budget constraint, whereby income can be expressed as the sum of consumption and savings (Heathcote et al. 2023, Fagereng et al. 2021).

$$Y = C + S$$

Moreover, I can further split consumption into food consumption and other types of consumption (durable and nondurable), therefore having:

$$Y = C_F + C_{NF} + S$$

Where C_F denotes food consumption, while C_{NF} denotes "other" consumption categories, not available in PLUS. Dividing both sides by Y and multiplying and dividing C_{NF} by C_F I get:

$$1 = \frac{C_F}{Y}\left[1 + \frac{C_{NF}}{C_F}\right] + \frac{S}{Y}$$

$$\frac{S}{Y} = 1 - \frac{C_F}{Y}\left[1 + \frac{C_{NF}}{C_F}\right]$$

Computing the difference in the saving rates for COL and JA workers using the above equation, I get:

$$\Delta\frac{S}{Y} \equiv \left(\frac{S}{Y}\right)^{JA} - \left(\frac{S}{Y}\right)^{COL} = -\Delta\left(\frac{C_F}{Y}\right)\frac{C}{C_F} = -\beta\frac{C}{C_F} \tag{1}$$

Equation (1) states that the difference in the saving rate equals the negative of the difference in the food share, multiplied by the ratio of total consumption to food consumption. The first term on the right-hand side is the estimated coefficient from a regression of the average propensity to consume food on the *Jobs Act* dummy. This coefficient captures how much less, on average, individuals hired under the JA spend on food compared to those hired under the

COL regime. The second term, by contrast, can be estimated from survey data. If one assumes homothetic preferences, the ratio of total consumption to food consumption should be the same across the entire income distribution. However, this assumption is unlikely to be true. Indeed, food is a necessary good and arguably constitutes a larger share of consumption for households at the bottom of the income distribution. Therefore, in applying (1), I divide the sample into deciles of income distribution and, for each decile, I apply the consumption ratios in Figure 3, which shows the average consumption ratio for each decile of the income distribution. More formally, let $z = 1, ..., 10$ denote the income decile to which households belong. The average change in the saving rate for households in decile z is then:

$$\Delta \left(\frac{S}{Y}\right)^{z} = -\beta \left(\frac{C}{C_F}\right)^{z}$$

Figure 3, based on the 2016 wave of the Bank of Italy *Survey of Household Income and Wealth* (SHIW), shows that the ratio of total to food consumption increases across the income distribution. It is around 2.5 in the first half, but rises significantly above 3 in the top two deciles.

4 Empirical Methodology

I exploit the discontinuity in employment protection introduced by the hiring date to estimate the impact of employment risk on consumption and hours worked. Specifically, I estimate the following equation by OLS for employees in large firms (i.e., firms with at least 15 employees) hired under open-ended contracts in 2018:

$$Y_i = \alpha + \beta \, JobsAct_i + \varphi \mathbf{X}_i + \gamma_{s(i)} + \varepsilon_i \tag{2}$$

In equation (2), the dependent variable Y_i is either the ratio of food consumption to family income (i.e., the average propensity to consume food) or its log. The vector $\mathbf{X}_i$ includes individual-level controls, including age (in bins), gender, marital status, family size, number of children, and education. $\gamma_{s(i)}$ represents sector fixed effects that control for hiring patterns that may vary across industries and for the unequal distribution of firms across sectors. Finally, ε_i is an idiosyncratic error term. I estimate two equations separately, like (2): one including only people who live in the North of Italy - the wealthiest region of the country -, and another only including people living in the Center and in the South - including Sicily and Sardinia. This is the reason why I do not include regional fixed effects.

The coefficient of interest is β, which captures the average difference in the outcome variable between workers hired before and after March 7, 2015. I expect the coefficient to be negative: higher employment risk should increase precautionary savings and thereby push consumption downward. Moreover, higher employment risk might reduce permanent income and, by the permanent income hypothesis, consumption. Causal identification relies on the assumption

that neither workers nor firms can manipulate hiring dates precisely around the cutoff, after controlling for demographics that may affect the hiring day. In Section 6, I provide evidence supporting this assumption.

Alternative Specification: Difference-in-Differences. On top of equation (2), I also consider a D-i-D strategy in which I exploit variation in both hiring date and firm size (i.e., above vs. below the 15-employee threshold) as in De Paola et al. (2021), and Bertoni et al. (2023). Table D1 in Appendix D presents the D-in-D estimates. D-in-D raises several concerns. First, marginal firms (e.g., those with 14 employees) might have strategically adjusted their size to benefit from the *Jobs Act*. The law stipulates that once a firm reaches 15 employees, newly hired workers are covered under the JA, as are those hired before March 7, 2015. Boeri and Garibaldi (2019) finds a large increase in firms with more than 15 employees after the reform, raising concerns about firms' selection at the dimensional threshold. Moreover, the reduction in firing costs was coupled with government hiring subsidies that presumably altered firms' hiring decisions (Boeri and Garibaldi 2019 and Sestito and Viviano 2018). Because small firms used the hiring subsidy disproportionately, this raises concerns about whether large- and small-firm workers are comparable in wages and the employment policies they face. Second, there are concerns related to self-selection in large or small firms based on consumers' attitude toward risk[7]. In particular, risk-averse individuals may prefer larger firms, creating a positive correlation between risk aversion and the D-i-D interaction term. This would lead to overestimating the magnitude of the effect[8]. Unfortunately, I can control for risk aversion only in 2018, since the question is asked only in the 2018 wave[9]. Third, and lastly, the data at my disposal do not allow me to test for pre-trends, since I have only one pre-treatment period.

5 Empirical Results

Food Consumption. Table 3 reports estimates for the average propensity to food consumption. Results are reported separately for people living in the North of Italy - the wealthiest region of the country - and the Center and the South (including Sicily and Sardinia). JA workers' average propensity to consume food is about 1.5 percentage points lower than that of COL workers in the North of Italy. In the South, however, the effect is not statistically significant. The intuition is as follows: people in the South are more likely to be liquidity-constrained.

[7]In a seminal paper, Guiso et al. (2005) show that firms pass through only a fraction of their idiosyncratic (permanent and transitory) shocks to wages. In this sense, they represent an important "vehicle of insurance provision" (cit. from the abstract). It's therefore sensible to imagine that risk-averse workers choose to work for the firm that will give them the best wage and employment insurance.

[8]The effect of risk aversion on current consumption is negative, through the Euler Equation, and the coefficient on the D-i-D interaction term is assumed to be negative. This means that the estimated causal effect is more negative, i.e., bigger in absolute value. Intuitively, more risk-averse people would consume less anyway and would select into bigger firms. Therefore, it's impossible to tell apart the effect of the JA on consumption.

[9]The real preference heterogeneity to control is prudence. However, if CRRA preferences are assumed, as stated in the literature review, the coefficient on relative prudence equals the coefficient on relative risk aversion plus 1.

Therefore, they save mainly to escape that situation and smooth consumption in the future. In other words, prudence doesn't play a major role in their saving decisions, and, at the margin, a change in employment risk does not induce them to consume less. Table B5 confirms this conjecture. It reports a regression of the probability of being HtM on a dummy equal to 1 for residing in the North and other controls. We define a household as HtM if its liquid wealth is below one-half of its monthly labor earnings. Columns 1 to 4, which report a linear probability model, show that people in the South and the Center are about 10 percentage points more likely to be HtM (both wealthy and poor), and therefore off the Euler Equation.

The results presented so far refer to the share of household income spent on food. Following Broda and Parker (2014), I rescale the estimated coefficient using equation (1) and Figure 3. In particular, I multiply the estimated coefficient by the ratio between total and food consumption for each decile of the net income distribution. The average increase in the saving rate is 4.24 percentage points, almost three times the effect on the average propensity to buy food. The reason for this higher effect (in magnitude) is that food is a necessary good. So people tend to cut less on food than on other consumption items, especially at the bottom of the distribution, where it represents a higher share of income.

Table 4 reports estimates using the log of the average propensity to food consumption as the dependent variable. Qualitatively, the results are similar: food consumption for JA workers is about 6% lower than for COL workers. This coefficient remains stable across all specifications.

BPP Imputation. To estimate the effect of the *Jobs Act* on a more comprehensive measure of consumption, that includes both durable and nondurable consumption, I follow the approach proposed by Blundell et al. (2008) and modified by Patterson (2023). More specifically, I estimate in HBS the following demand for food spending, available both in HBS and PLUS:

$$\log(f_i) = \alpha + \mu \mathbf{X}_i + \delta \log(c_i) + \varepsilon_i \tag{3}$$

In the above equation, f_i is food spending, $\mathbf{X}_i$ is a vector of demographic controls available in both surveys[10]. δ is the elasticity of food consumption with respect to total spending (durable and nondurable). The error term captures both measurement error in food consumption and unobserved heterogeneity in food demand (Blundell et al. 2008). I invert this demand function to back out an imputed measure of spending in PLUS using the coefficients estimated by OLS. Panel A and Panel B of Table 5 show the estimated coefficient of equation (2) for the (imputed) log of consumption and the average propensity to consume, respectively. Both dependent variables are multiplied by 100. The first three columns report estimation results for large firm workers without sector fixed effects (column 1), with job sector fixed effects (column 2), and with risk aversion dummies (column 3). The last three columns report the same specifications for small firm employees. The JA's effect on imputed spending is around 8%. This estimate is higher in magnitude than the baseline on the log of food consumption. This result is consistent

[10]The vector includes: age bins, gender, place of residence, Italian citizenship, education levels, marital status, and family size.

with the fact that food consumption is a necessary good, and so people cut it proportionately less than other non-essential consumption items.

Appendix C shows the details of the imputation. Table C1 shows the summary statistics of HBS, while Table C2 shows the results of the imputation. Figure C1 compares the two distributions. Both have the same mean (around 7.7), but the imputed distribution has fatter tails. Moreover, Figure C2 (left panel) shows a strong correlation between the imputed and actual measures: the coefficient of a regression of one measure on the other, without the constant, is very close to 1, with an R^2 of 0.98. The right panel of Figure C2, by contrast, shows the scatter plot and confirms that the imputed distribution has fatter tails than the actual one, as many bin points lie far from the 45-degree line.

The Effect on Labor Supply. Table B2 shows results for the intensive margin of labor supply, defined as the average weekly hours worked. While the first two columns suggest an increase in weekly hours worked, the effect disappears once one controls for tenure. Indeed, whether by controlling for tenure or by keeping only individuals with maximum years of tenure, the effect is no longer distinguishable from 0 economically or statistically.

6 Placebos and Robustness Checks

In this section, I perform a battery of placebo tests and robustness checks to strengthen the causal claim made above. I also examine how the estimated causal effect varies across heterogeneity dimensions.

Income Differences. I first examine whether the *Jobs Act* is correlated with income. It is important to confirm that incomes did not change as a result of the reform. Figure 4 plots the estimated coefficients from a regression of income (in both levels and logs) on the *Jobs Act* dummy, controlling for the same covariates used in the main analysis. The 95% confidence intervals are also displayed. In both cases, the coefficients are neither statistically nor economically different from zero. Of course, as discussed in the model section, this does not imply that the present discounted value (PDV) of income is the same for JA and COL workers. Since JA workers face a higher probability of dismissal, they receive unemployment insurance more frequently, leading to lower cash flows to be discounted at the same interest rate. Nonetheless, it is important to establish *prima facie* that there are no systematic differences in earnings (or productivity) between JA and COL workers. Two reasons are behind this result. First, the Italian labor market is characterized by a high degree of centralization of bargaining. Wages are set by agreements between trade unions and employers, and there is little scope for decentralized wage-setting. The second reason is that since the Jobs Act did not change the contracts, as highlighted in Section 2, there is no reason, a priori, for firms to pay a wage differential to JA to compensate for higher employment risk.

The similarity of income between JA and COL workers might, at first, seem at odds with

the (slight and overall not significant) increase in weekly hours worked found in Table B2. However, the survey question on weekly working hours does not distinguish between paid and unpaid work. For this reason, people may be inclined to do some unpaid work to "please" the boss and reduce employment risk. This might explain why people work more but do not earn more.

Small Firms and Public Employees. As explained in Section 2, the *Jobs Act* changed the way individual unlawful dismissals are regulated for employees of large firms. However, it did not affect the regulation for employees of small firms, employing fewer than 15 workers. Therefore, the latter ones represent a natural placebo group. I estimate the same specification as in equation (2) for small-firm employees. Since nothing changed in individual employment protection for employees of firms with fewer than 15 workers, I should observe no difference in consumption between workers hired after and before March 7, 2015. Table 6 shows that the estimates are statistically and economically indistinguishable from zero. The last three columns of Table 5 confirm this result, where only small-firm workers are considered. An additional placebo test verifies that there is no effect for public sector employees, since the *Jobs Act* does not apply to them. I therefore estimate equation (2) for public employees only. Table 7 reports results for the food share and the log of food consumption under the same specifications as before. The coefficient is both economically and statistically indistinguishable from zero.

Bigger Firms. Following De Paola et al. (2021), I restrict the sample to employees in firms with at least 20 workers and re-estimate the main equations for both food consumption and labor supply. While firms slightly below the 15-worker threshold may have some incentive to hire two or three additional workers to benefit from lower firing costs, firms well above the threshold certainly do not. The estimated coefficients (in Table 8) are essentially unchanged and remain statistically significant. This further supports the conclusion that firm size self-selection is not driving the results. Figure 7 shows the effect including only workers of firms with 20+, 30+ and 40+ employees.

Hiring Rates. A final concern, connected to the one raised above, is whether firms altered their hiring behavior around the time of the reform because they were aware of it. Although a reform aimed at reducing labor market rigidity and duality had long been anticipated, firms or workers likely could not have foreseen the specific discontinuous change in employment protection implemented on March 7, 2015. Additionally, the law was published in the *Gazzetta Ufficiale* on March 6, 2015, and took effect the very next day, leaving neither workers nor firms time to adjust. Figure 5, produced using data sourced from Italy's *Labor Force Survey*, shows the share of workers in the sample hired in the quarters before and after the first quarter of 2015 ($t = 0$). The hiring rate appears unchanged around this period, and the distribution is fairly uniform for both large and small firms. Figure 6 provides additional evidence by plotting

the monthly growth rate of employment from March 2012 to March 2018. The average growth rate before the reform was 0.02%, compared to 0.06% after the reform. This change is not statistically different from 0 because the time series is highly volatile. This last finding aligns with Fana et al. (2015), which argues that the JA did not significantly change employment growth.

Do workers go to the small firms after the introduction of the Jobs Act? An additional concern is that workers might move to small firms not involved in the reform to get higher employment protection. I believe that this concern is misplaced for two reasons. First, big firms' JA workers, despite having lower protection than their COL colleagues, are still more protected than small firms' workers, to whom neither the COL nor the JA applies. The second reason is that Figure 5 shows no jump in small firms' hiring rate, suggesting no significant increase in hiring after the reform.

Risk Aversion and Tenure. As anticipated in Section 2, the 2018 survey includes questions aimed at eliciting individual discount factors and relative risk aversion, allowing me to control for preference heterogeneity (Jappelli and Pistaferri 2020). The wording of these questions is provided in Appendix A. The first question identifies the discount factor, assuming it equals one over one plus the interest rate. The second question elicits the certainty equivalent, allowing one to infer the coefficient of relative risk aversion under CRRA utility. I estimate the following specification:

$$Y_i = \alpha + \beta\, JobsAct_i + \varphi \mathbf{X}_i + \theta_{s(i)} + \gamma_{\tau(i)} + \sigma_{r(i)} + \rho_{d(i)} + \varepsilon_i \tag{4}$$

In equation (4), $\gamma_{\tau(i)}$, $\sigma_{r(i)}$, and $\rho_{d(i)}$ are dummies representing different levels of tenure, risk aversion, and discount factor, respectively. The tenure FEs are 5-year bins. Table 9 shows the estimated coefficients of (4). As columns 1 and 4 suggest, controlling for preference heterogeneity makes little difference in the estimated coefficient, both for the level of the food ratio and its log. This is reassuring, as workers do not self-select into or out of the *Jobs Act* based on their preferences. Controlling for tenure, by contrast, increases the estimated effect by about one percentage point: to 2.61 from 1.51. Finally, restricting the sample to individuals with at most 5 years of tenure, despite the significant reduction in sample size, does not affect the results.

Heterogeneity by Age. The first dimension of heterogeneity I explore is age. Gourinchas and Parker (2002) argue that households transition from buffer-stock saving behavior (Carroll 1997) to certainty-equivalent consumption plans (Hall 1978) around the age of 40–45. As a result, income risk should matter less—or not at all—for older individuals. The empirical findings (in Table 10, columns 1 and 2) are consistent with this prediction: the reduction in consumption is statistically significant only for individuals under 40, while it is indistinguishable from zero for those aged 40 and above.

Heterogeneity by education and income. The next dimension I look at is education. Columns 3 and 4 of Table 10 suggest the effect is higher for people with a college degree. This is because higher education typically leads to higher and more stable incomes (Savoia 2023). Therefore, on the margin, a reduction in employment risk matters more for the top of the income distribution. To dig deeper into this potential explanation, columns 5 and 6 split the sample according to whether income is below or above the median. The effect is stronger for above-median incomes, supporting the idea that better-educated individuals have more to lose from the *Jobs Act*.

Union Representation. The PLUS dataset also asks respondents whether their workplace has union representation. The effect is significant only for unionized firms (Table 10, columns 7 and 8). This result speaks to the literature investigating the importance of union representation at the firm level as an insurance mechanism. Kim et al. (2018) shows that union representation in the workplace fosters risk-sharing and protects workers against adverse shocks. It might be that the JA reduced, within the workplace, unions' capacity to protect workers, maybe because it altered the relationship between employees and employers in a more favorable way to the latter.

7 The Model

Why a model? As mentioned in the introduction, higher employment risk can affect consumption via two channels. The first channel is a stronger precautionary saving motive (Blanchard and Mankiw 1988, Kimball 1990). More volatile income - induced by the JA - leads consumers to save more to insure against negative employment shocks. Furthermore, a higher separation rate lowers the PDV of income, reducing consumption as predicted by the standard PIH (Hall 1978, Friedman 1957). Because both channels drive consumption down, and because I cannot observe the PDV in PLUS, I cannot disentangle the two channels in the empirical analysis above. This point is even more pressing given that I do not observe wealth in PLUS (only a proxy for it). From an econometric point of view, in the above empirical part, I essentially estimate this reduced form model (with no controls for expositional convenience):

$$\log(C_{it}) = \alpha + \beta JA_{it} + \varepsilon_{it} \tag{5}$$

However, the true model is likely to resemble the following equation more closely:

$$\log(C_{it}) = \alpha + \beta JA_{it} + \vartheta PDV_i + \delta A_{it} + \xi_{it} \tag{6}$$

Where PDV_i is the present discounted value of income and A_{it} is assets. A straightforward

application of OLS algebra (Angrist and Pischke 2009) yields:

$$\hat{\beta} = \beta + \frac{Cov\left[\vartheta PDV_i + \delta A_{it}, JA_{it}\right]}{Var\left(JA_{it}\right)}$$

$$= \underbrace{\beta}_{\text{True effect of Employment Risk}} + \underbrace{\frac{Cov\left[\vartheta PDV_i + \delta A_{it}, JA_{it}\right]}{Var\left(JA_{it}\right)}}_{\text{PIH Channel}}$$

The formula above shows that the OLS coefficient estimated above, even though causally identified, bundles the two components. It also conveys an additional insight: since PDV and A affect consumption positively (ϑ and δ are both positive), and the *Jobs Act* affects both the PDV and wealth negatively, the estimated reduced-form effect is an upper bound (in magnitude) for the *sole* risk effect of risk on consumption. This makes the decomposition more important, because in principle the effect of risk could be wiped out once controlling for permanent income. To quantify each component's contribution to the total effect, I therefore rely on the following model.

Another shortcoming of the reduced-form approach is that I cannot say much about the structural parameters that underpin the effect of the JA on consumption. For this reason, I estimate the structural preference parameters with MSM, using the empirical estimate presented above as a target moment (together with other moments from the wealth distribution and the labor market). This bridges the three strands of the literature mentioned in the literature review. The approach of estimating structural models starting from empirically identified causal effects has been gaining popularity, as the recent works by Daminato and Padula (2024) and Choukhmane and De Silva (2026) - among others - show.

Model Setup. Time is discrete, and the horizon is infinite. Each period is one year. There is no aggregate risk and no mortality risk. This assumption is particularly crucial in my setting because it rules out layoffs, which are not covered by the *Jobs Act* regulation[11]. The economy has two types of consumers/workers: a share $1-\omega$ of *Chart of Labor* workers and a share ω of *Jobs Act* workers. Their preferences are defined by a CRRA utility function with the same relative risk aversion coefficient γ[12]. They value the future at the discount factor β. At each point in time, they draw their idiosyncratic productivity from the same stochastic process defined below. They face two different employment risks, however. Employment transitions for COLs are regulated by the matrix Π_{COL}, with job-finding probability f and separation rate s_{COL}. The matrix for JAs is, instead, Π_{JA} with the same f but with $s_{JA} > s_{COL}$. The "outer" employment transition matrix governing employment transitions and the matrix governing productivity draws are independent. This means that employees face the same unemployment risk regardless of their productivity draw. This rules out employer decisions to fire workers

[11] There were reforms to the regulation of mass-layoff procedures, but they apply regardless of the hiring date. So, they do not interact with the JA.

[12] In Appendix F1, show that results are very close to the baseline with Epstein-Zin preferences.

being contingent on productivity. I make this assumption to isolate employment shocks as much as possible, holding other sources of risk fixed.

$$\max_{\{a_{j,t+1},c_{j,t}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{c_{j,t}^{1-\gamma} - 1}{1-\gamma} \right]$$

$$a_{j,t+1} = (1+r)a_{j,t} + y\big(e_t(e_{t-1}, \Pi_j), p_t\big) - c_{j,t}$$

$$a_{j,t+1} \geq 0$$

$$y\big(e_t(e_{t-1}, \Pi_j), p_t\big) = \begin{cases} w \times p_t & \text{if} \quad e_t = 1 \\ b(p_t) & \text{if} \quad e_t = 0 \end{cases} \tag{7}$$

$$b(p_t) = \min\Big\{ b_0 \times p_t, b_1 \times p_{50} \Big\}$$

$$\log(p_t) = \rho \log(p_{t-1}) + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma_\varepsilon^2)$$

$$\Pi_j = \begin{bmatrix} 1-f & f \\ s_j & 1-s_j \end{bmatrix}, \quad j \in \{COL, JA\}$$

When employed ($e_t = 1$), workers earn their productivity draw times w (normalized to 1); when unemployed ($e_t = 0$), their productivity draw times $b(p_t)$, i.e. the replacement rate of the unemployment insurance. Unemployment insurance equals 0.75 of productivity ($b_0 = 0.75$) if productivity is below the cross-sectional median (p_{50}). Otherwise, it's capped at slightly more than half of median productivity ($b_1 = 0.54$). This design is meant to replicate as closely as possible the Italian regulation of unemployment benefits. These parameters are type-invariant, meaning that they are the same for JA and COL workers. In the optimal unemployment insurance exercise, I look for potentially type-variant parameters that maximize welfare.

Idiosyncratic productivity is modeled as standard: its log $\log(p_t)$ follows an AR(1) with persistence ρ and variance of the stochastic component σ_ϵ^2. Consumers can save in a liquid asset that gives a fixed return of r and cannot borrow. They choose consumption each period to maximize the discounted sum of expected utility, subject to the dynamic budget constraint and the borrowing constraint, respectively, the second and third lines of equation (7).

Calibration and Estimation. The model's calibration proceeds in two steps. First, I selected a set of parameters externally, using values commonly reported in the literature. I then estimate a second set of parameters internally with indirect inference methods. I directly assign the parameters of the stochastic productivity process, the borrowing constraint, the interest rate, the wage, and the shares of COL and JA workers. I model the productivity process with an AR(1) with 7 states. The persistence parameter ρ is set to 0.95, while the standard deviation of the productivity is $\sigma_\varepsilon = 0.5$ (from Jappelli et al. 2024. We normalize the wage w to 1, and set the interest rate to 2% annually. We set the borrowing constraint to 0.

The share of JAs ω_{JA} is 0.17, matching the share of large-firm workers hired after March 7, 2015, as of 2018.

I calibrate five parameters internally using the Method of Simulated Moments (MSM). I choose the parameters of interest to minimize the distance between key model-implied moments and their empirical counterpart. This amounts to solving the following minimization problem. Let Θ and $\hat{\Theta}$ denote the vectors of the true and estimated parameters, respectively. Let $\mathbf{W}$ be a weighting matrix and ψ the vector of the externally set parameters. Let $g(\Theta;\psi)$ be the vector of theoretical moments and m the vector of data moments. The minimization problem can be written as:

$$\hat{\Theta} = \arg\min_{\Theta} \Big[g(\Theta;\psi) - m\Big]' \mathbf{W} \Big[g(\Theta;\psi) - m\Big] \tag{8}$$

$$\Theta = \Big[\gamma, \beta, s_{COL}, s_{JA}, f\Big]$$

Details of the inference are in Appendix F5. I estimate the following four parameters with MSM: the relative risk aversion parameter γ, the discount factor β, the separation rates for COLs and JAs (s_{COL} and s_{JA}), and the job-finding rate f. I also impose that $s_{JA} > s_{COL}$. On top of the difference in consumption between JAs and COLs found in the empirical part (-7.91%), I target the following moments: the average wealth-to-income ratio of Italian households whose head is aged between 25 and 64, its median, and the P90, sourced from the 2016 SHIW; the average tenure of COL workers of years, the average and the long-term unemployment rate (i.e., people who have been in unemployment for more than one period in the model) between 2016 and 2019, respectively. I source the moments of the cross-sectional distribution of the wealth-to-income ratio from the 2016 SHIW and, following Jappelli and Pistaferri (2014), define wealth as the sum of real and financial assets minus unsecured debt, while income is net family income. I obtain the average COL tenure from the 2018 quarterly waves of the LFS, and I use aggregate unemployment statistics from the Federal Reserve Bank of St. Louis website and EUROSTAT for the remaining two targeted moments (details in Table 12). While the parameters are jointly estimated, single moments are more informative about single parameters. The empirical difference in consumption between COL and JAs is meant to inform risk aversion. The moments of the wealth-to-income ratio distribution are particularly informative on the discount factor β, and the average tenure of COL workers informs s_{COL}. In contrast, the total and long-run unemployment rates are more informative about s_{JA} and f.

Table 11 shows the calibration for the assigned parameters. The first panel of Table 12, by contrast, shows the MSM's performance by comparing targeted empirical moments with their theoretical counterparts. The second panel shows the estimated parameters with their standard errors. Overall, the model matches the targeted parameters fairly closely. Relative risk aversion is estimated at around 1.5. This value falls within the range of estimates in the papers mentioned above (Bertola et al. 2005, Guiso and Jappelli 2024a, Sciacchetano (2024), Christelis

et al. 2020). The discount factor β is 0.97. At the same time, the separation rate for COLs is 4.7%, close to the separation rate found (transitions from Employment to Unemployment) by D'Amuri et al. (2022), who find it to be around 4.8% (annualized) in the period before the financial crisis. The separation rate for JAs is around 13%, while the finding rate is 44%. The standard errors of the MSM (shown in parentheses in Table 12) are very low, suggesting precise estimates. The first takeaway is that the consumption differences between JA and COL workers can be explained by a standard incomplete model with ex-ante heterogeneity in employment risk and reasonable preference parameters. What remains to be discovered, however, is how much of this consumption wedge is explained directly by higher employment risk exposure and how much by differences in permanent income.

Model Regression Results. Figure 8 shows the consumption policy function for JA (red line) and COL (blue line) in the states of employment (left panel) and unemployment (right panel), both in the median productivity state. JA workers' consumption - conditioning on the same wealth, productivity, and employment status - is always below that of COL workers. In other words, JA workers choose to consume less than COL workers, even when they have the same resources. Following the above discussion, this result might be either due to higher precautionary saving or a lower permanent income.[13]

Figure 9, on the other hand, shows the distribution of consumption (left panel) and the present discounted value of income (right panel) for JA and COL workers, in red and blue, respectively. The PDV of income is calculated by summing all income an individual earns over the simulated period, discounted by $(1 + r)$ to the power of the period.[14]. The COLs almost first-order stochastically dominate the JAs for both consumption and PDV. This finally confirms that the concerns raised at the beginning of this section were well posed. To see which channel prevails, I estimate (5) and (6) with simulated data and report the results in Table 13. Column 1 shows the mean difference in the log of consumption between JA and COL workers. This difference is 7.9%, the value targeted in the risk-aversion estimation. Column 2 controls for the PDV of income and wealth. I find that the coefficient on *Jobs Act* is 2.7%. This means that the sole effect of employment risk accounts for roughly one-third of the total effect of the *Jobs Act*. Column 3 allows for a more flexible specification with the interaction between employment risk, PDV, and assets, finding that the effect of the JA decreases in magnitude as the PDV and assets increase. This is intuitive: as a consumer has more cash on hand, she can smooth consumption, and the precautionary saving motive weakens (i.e., she is on the "linear" side of the consumption policy function).

The importance of the precautionary saving motive due to the JA also appears when I regress wealth (scaled by income) on the JA dummy. This specification resembles Fuchs-Schündeln and Schündeln (2005). I do this exercise in columns 4 and 5 of Table 13. When I do not control

[13] Figure B2 plots the mean income for JA and COL workers over the simulated period of the model.

[14] $PDV_i = \sum_{t=1}^{200} y_{it} \times (1 + r)^{-(t-1)}$

for income - i.e., I look at the unconditional mean difference in assets between JAs and COLs - the coefficient is negative, paradoxically suggesting that JA workers save less than COLs. This is, however, because JAs spend, on average, more time unemployed and hence have lower lifetime income. Indeed, when I control for the PDV, and so I condition on the same amount of permanent resources, the difference in savings turns significantly positive.

Model Extensions. Appendix F2 and Appendix F3 proposes two variants of the baseline model. The first variant uses assumed preferences: recursive preferences à la Epstein and Zin (1989) instead of CRRA. With recursive preferences, two distinct parameters govern risk aversion and the elasticity of intertemporal substitution. The first parameter governs individuals' willingness to smooth consumption over time, whereas the second parameter governs their willingness to smooth consumption across states. I estimate the risk aversion to be 1.74, the EIS 0.30, and the discount factor 0.95, and the fit of the model is remarkably good (Table F1). A comparison between Table F2 and Table 13, finally, shows no economically meaningful differences between the two model specifications.

The second extension I propose is to introduce a second, illiquid account as in Auclert et al. (2024), Kaplan and Violante (2022), and Graves (2025). In this more realistic model, consumers can save in a liquid asset and an illiquid asset. The latter pays a higher return, but can be accessed each period with an iid probability χ[15]. Consumers maximize the same utility function - assumed to be CRRA -, but face the following constraints[16]:

$$\begin{aligned} c_t + b_{t+1} + d_t \, \mathbb{I}_{adjust=1} &= (1 + r_b) b_t + y\big(e_t(e_{t-1}, \Pi_j), p_t\big) \\ a_{t+1} &= (1 + r_a) a_t + d_t \, \mathbb{I}_{adjust=1} \end{aligned} \tag{9}$$

When consumers can adjust the illiquid account ($\mathbb{I}_{adjust=1}$), they deposit an amount d_t into the illiquid account. If adjustment is not possible ($\mathbb{I}_{adjust=0}$), the problem is the same as in the one-asset case. Neither the liquid nor the illiquid account can be shorted. Appendix F contains the recursive formulation of the model and calibration details.

The insight from the papers mentioned above is that people save in the liquid account to keep cash available in case a temporary negative shock occurs. The illiquid account, by contrast, is used only for intertemporal saving and is withdrawn only in response to large shocks. In particular, Graves (2025) shows that when the separation rate increases, households increase their holdings in the liquid account to buffer more wealth against employment risk[17]. In my setting, I should expect JAs workers to hold a higher share of their wealth in the illiquid account, as the risk of becoming unemployed is larger for them. This is exactly what I found in the model. While the difference in consumption is close to the one-asset model, JAs workers

[15] As argued by Auclert et al. (2024), this feature produces some form of transaction costs attached to the illiquid account in a way that is less computationally intensive than calibrating a transaction cost function directly.

[16] I omit the j index in the budget constraint.

[17] Table F3 shows the estimated parameters with the targets.

have a liquid-wealth-to-income ratio about 20% higher than COLs, as shown in Table F4.

Life Cycle Version. Additionally, Appendix F4 shows a life cycle version of the model, in which all individuals start working with the COL but, upon finding a new job, they transition to the JA. Unlike the baseline model, where belonging to either the COL or the JA regulation is set ex ante, here COL or JA membership is an absorbing state. I estimate the model's risk aversion and discount factor by targeting the working-age profile of the wealth-to-income ratio (ages 25 to 65). The estimated risk aversion in the life cycle version is 1.48, while the discount factor is 0.99, surprisingly close to the estimates in Daminato and Padula (2024), who estimate the two parameters to be 1.61 and 0.99, respectively.

8 Optimal Unemployment Insurance

In the model above, unemployed workers receive means-tested UI benefits. However, the formula used to compute the benefits is the same for both JA and COL workers. In this final part of the paper, I ask the following questions. How should policymakers account for this duality in the labor market when designing unemployment insurance benefits? In the spirit of Pellegrini (2025) and Zanella Cavallero et al. (2026), I use the model to find the optimal UI structure for JA and COL workers that maximizes overall welfare, subject to an invariant fiscal cost. In practice, I solve this maximization problem:

$$\max_{\Omega} (1-\omega_{JA}) \int V_{COL}(S;\Omega) g_{COL}(S;\Omega) dS + \omega_{JA} \int V_{JA}(S;\Omega) g_{JA}(S;\Omega) dS$$

s.t.

$$(1-\omega_{JA}) \left[(1-empl_{t,COL}) \left(\int_{p_t} \min\left\{ b_{0,COL} p_t ; b_{1,COL} p_{50} \right\} d\mu \right) \right] + \tag{10}$$

$$\omega_{JA} \left[(1-empl_{t,JA}) \left(\int_{p_t} \min\left\{ b_{0,JA} p_t ; b_{1,JA} p_{50} \right\} d\mu \right) \right] \leq \overline{T}(\Theta, \psi)$$

$$\Omega \equiv \{ b_{0,COL}, b_{1,COL}, b_{0,JA}, b_{1,JA} \}$$

In (10), I maximize over the parameters used to compute the type-specific UI:

$$b_{0,COL}, b_{1,COL}, b_{0,JA}, b_{1,JA}$$

S is the state space, made of three dimensions: employment status, productivity, and wealth. In the integrals, g denotes the distribution over all states. At the same time, μ is the distribution over productivity[18]. $\overline{T}$ (that is, a function of Θ and ψ) is the amount of money spent in the

[18]Since employment and productivity states are independent, the productivity distribution for JA and COL

baseline regime. The left-hand side of the budget constraint is the amount of money the Government Spends to provide "tailored" UI to JAs and COLs. Table 14 shows the results. The optimal UI parameters for COLs are lower than the ones for JAs. This suggests that, to maximize welfare, the Government should provide more generous insurance to the latter compared to the former, since they are more exposed to employment risk.

One caveat applies: in this model, workers do not decide if and how much to work, as an exogenous process regulates employment transitions and labor is supplied inelastically. This means the model cannot capture household moral hazard. Including moral hazard would certainly change the model's conclusions, since more or less generous UI benefits influence unemployed people's effort in job search (Lalive et al. 2015 and Landais 2015).

Table 15 replicates the baseline results using the UI parameters obtained in (10) and displayed in Table 14. The unconditional difference in consumption goes down to roughly 6%. In other words, providing more insurance reduces the consumption gap between JAs and COLS by almost 2 percentage points.

9 Conclusion and Direction for Further Research

In this paper, I leverage the Italian *Jobs Act* (JA) to investigate the relationship between employment risk, consumption, and labor. I find that workers affected by the reform reduce their food spending-to-income ratio by 1.5 percentage points, corresponding to an 8% drop in spending and a 4-percentage-point increase in the saving rate. These effects are stronger among younger and more-educated workers in Northern Italy.

I then develop a Bewley-Huggett-Aiyagari model with ex-ante heterogeneity in employment risk. After calibrating the model's key parameters to match the empirical effect mentioned above, I use it for two purposes. First, I use it to disentangle the pure effect of employment risk from the permanent income hypothesis (PIH) channel, which arises from a lower present discounted value (PDV) of income. Running a regression on simulated data, controlling for PDV and asset levels, I find that employment risk heterogeneity accounts for about one third of the model-implied consumption difference.

Second, I look for the optimal, potentially type-dependent, structure of unemployment insurance that maximizes welfare, under a constant government budget. I find that, to maximize welfare, the Government should provide more insurance to JAs and less to COLs than in the baseline model, where UI is the same for both types of workers. This not only maximizes welfare, but also reduces the consumption inequality by almost 2 percentage points. This result comes with the caveat that employment decisions are exogenous in the model, and so they are free of moral hazard implications.

This paper carries two policy-relevant messages. First, supply-side structural reforms might spill over into aggregate demand, affecting households' consumption decisions. Second, the *Jobs Act* created an additional layer of job market duality within open-ended employees - on

is the same.

top of the existing one between fixed-term and open-ended contracts - that fuels consumption inequality. This link between employment protection and consumption decisions may be highly relevant to the effectiveness of monetary and fiscal policy.

This work can be improved along several dimensions. To the best of my knowledge, currently, no dataset provides joint information on firm size, employment protection, and detailed spending categories beyond food. While the results presented here are informative and consistent with theoretical predictions - further validated by model simulations - access to actual (e.g., self-reported) consumption data, rather than imputed, and possibly wealth, would represent a substantial data improvement. Finally, a comprehensive welfare evaluation of the JA should combine the demand-side effects examined here (i.e., the impact on the precautionary saving motive) with the supply-side impacts that the reform aimed to achieve.

Finally, recent literature (see Kaas et al. 2026, Pellegrini 2025 and Zanella Cavallero et al. 2026) has highlighted how wealth heterogeneity, induced by different careers and income histories, shapes job mobility. Further research should focus on the opposite direction: how job market frictions and regulations shape consumption and portfolio choices.

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Tables

Table 1: Summary Statistics by firm size

	Large Firms		Small Firms		All	
	Mean	Sd	Mean	Sd	Mean	Sd
Between 18 and 24	0.03	0.16	0.04	0.20	0.03	0.17
Between 25 and 29	0.06	0.24	0.09	0.28	0.07	0.26
Between 30 and 39	0.26	0.44	0.29	0.45	0.27	0.44
Between 40 and 49	0.37	0.48	0.34	0.48	0.36	0.48
Between 50 and 64	0.28	0.45	0.24	0.43	0.27	0.44
Male	0.65	0.48	0.51	0.50	0.61	0.49
Married	0.62	0.49	0.58	0.49	0.60	0.49
Family Size	3.02	1.18	3.05	1.19	3.03	1.18
N. of Children	1.10	1.02	1.05	1.01	1.08	1.02
South and Islands	0.17	0.38	0.21	0.41	0.19	0.39
Centre	0.21	0.40	0.23	0.42	0.21	0.41
North	0.62	0.49	0.56	0.50	0.60	0.49
Low Education	0.36	0.48	0.43	0.50	0.38	0.49
Middle Education	0.49	0.50	0.48	0.50	0.49	0.50
High Education	0.15	0.35	0.09	0.28	0.13	0.33
Union Representation	0.69	0.46	0.05	0.21	0.47	0.50
Jobs Act	0.06	0.25	0.09	0.28	0.07	0.26
Food Share (%)	29.37	12.25	30.29	12.76	29.66	12.42
Food Consumption (€100)	6.72	4.01	6.16	3.86	6.55	3.97
Individual Net Income (€100)	13.97	5.23	10.94	4.47	13.01	5.19
Family Net Income (€100)	23.21	10.63	20.31	9.74	22.29	10.44
Weekly average hours worked	37.47	8.26	34.63	10.76	36.55	9.24
Employed in a large Firm	·	·	·	·	0.68	0.47

Notes: This Table displays the summary statistics for the 2014 and 2018 PLUS waves by firm size.

Table 2: Summary Statistics by contract type - only large firm employees

	JA		COL	
	Mean	Sd	Mean	Sd
Between 18 and 24	0.12	0.33	0.02	0.14
Between 25 and 29	0.20	0.40	0.05	0.23
Between 30 and 39	0.31	0.46	0.26	0.44
Between 40 and 49	0.24	0.43	0.37	0.48
Between 50 and 64	0.13	0.34	0.29	0.45
Male	0.63	0.48	0.66	0.48
Married	0.41	0.49	0.63	0.48
Family Size	2.99	1.22	3.02	1.18
N. of Children	0.71	0.98	1.12	1.02
South and Islands	0.20	0.40	0.17	0.38
Centre	0.20	0.40	0.21	0.41
North	0.60	0.49	0.62	0.48
Low Education	0.23	0.42	0.37	0.48
Middle Education	0.52	0.50	0.49	0.50
High Education	0.26	0.44	0.14	0.35
Union Representation	0.46	0.50	0.70	0.46
Food Share (%)	27.14	12.81	29.55	12.19
Food Consumption (€100)	6.49	4.12	6.74	4.00
Individual Net Income (€100)	13.43	5.33	14.01	5.22
Family Net Income (€100)	24.42	10.69	23.13	10.62
Weekly average hours worked	38.24	9.05	37.42	8.20

Notes: This Table displays the summary statistics for the 2014 and 2018 PLUS waves by contract type (JA and COL). Only large firm employees are included. Sample weights are used.

Table 3: Food Share (× 100)

	North			Centre and South		
	(1)	(2)	(3)	(4)	(5)	(6)
Jobs Act	-1.63**	-1.72**	-1.57**	0.74	0.59	0.72
	(0.68)	(0.69)	(0.68)	(1.03)	(1.05)	(1.04)
Male	-1.59***	-1.53***	-1.14**	-1.66**	-1.45**	-1.33*
	(0.51)	(0.54)	(0.53)	(0.68)	(0.70)	(0.69)
Married	-1.26**	-1.18*	-0.44	-0.70	-0.64	-0.01
	(0.62)	(0.63)	(0.62)	(0.87)	(0.88)	(0.87)
Family Size	1.94***	1.89***	2.82***	2.36***	2.37***	2.97***
	(0.31)	(0.32)	(0.33)	(0.34)	(0.35)	(0.36)
N. of Children	0.91**	0.94**	0.34	-0.22	-0.28	-0.55
	(0.39)	(0.39)	(0.40)	(0.48)	(0.48)	(0.48)
Between 25 and 29	-2.51***	-2.57***	-3.41***	-2.93**	-3.46***	-4.56***
	(0.97)	(0.97)	(0.95)	(1.29)	(1.32)	(1.33)
Between 30 and 39	-1.02	-1.07	-1.64**	-1.62*	-1.88**	-2.63***
	(0.69)	(0.70)	(0.69)	(0.90)	(0.92)	(0.92)
Between 40 and 49	-0.81	-0.90	-1.17*	-1.19	-1.37	-2.11**
	(0.65)	(0.66)	(0.65)	(0.84)	(0.85)	(0.84)
Years of Education	-0.58***	-0.55***	-0.33***	-0.74***	-0.68***	-0.43***
	(0.08)	(0.08)	(0.09)	(0.10)	(0.10)	(0.10)
Net Family Income (€100)			-0.23***			-0.23***
			(0.03)			(0.03)
Observations	2227	2227	2227	1461	1461	1461
Controls	✓	✓	✓	✓	✓	✓
Job Sector FE		✓	✓		✓	✓

Notes: This Table shows OLS estimates of (2). The dependent variable is food consumption as a share of family income. Jobs Act is a dummy equal to 1 if the worker is hired after March 7, 2015. The sample includes only large firm workers in 2018. Standard errors, clustered at the individual level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 4: Log of Food Share (× 100)

	North			Centre and South		
	(1)	(2)	(3)	(4)	(5)	(6)
Jobs Act	-6.25**	-6.58**	-5.92**	-0.34	-0.89	-0.70
	(2.87)	(2.89)	(2.88)	(3.75)	(3.81)	(3.76)
Male	-6.57***	-6.09***	-4.70**	-4.57*	-3.64	-3.28
	(1.96)	(2.10)	(2.07)	(2.52)	(2.64)	(2.60)
Married	-5.46**	-5.04**	-1.95	-2.35	-1.99	0.62
	(2.43)	(2.43)	(2.43)	(3.21)	(3.23)	(3.16)
Family Size	7.09***	6.89***	10.38***	8.99***	8.96***	11.46***
	(1.25)	(1.25)	(1.31)	(1.32)	(1.34)	(1.39)
N. of Children	3.96***	4.07***	1.80	-0.70	-0.93	-2.10
	(1.45)	(1.44)	(1.45)	(1.73)	(1.73)	(1.72)
Between 25 and 29	-11.55***	-11.83***	-14.82***	-10.34**	-12.03**	-15.98***
	(4.17)	(4.18)	(4.11)	(4.86)	(4.94)	(4.93)
Between 30 and 39	-3.00	-3.27	-5.28**	-4.60	-5.42	-8.20**
	(2.63)	(2.65)	(2.64)	(3.27)	(3.33)	(3.32)
Between 40 and 49	-4.10*	-4.43*	-5.35**	-4.49	-5.12*	-7.82**
	(2.48)	(2.49)	(2.47)	(3.06)	(3.10)	(3.05)
Years of Education	-2.43***	-2.32***	-1.52***	-2.66***	-2.50***	-1.58***
	(0.30)	(0.32)	(0.33)	(0.36)	(0.37)	(0.39)
Net Family Income (Log x 100)			-0.21***			-0.22***
			(0.03)			(0.03)
Observations	2227	2227	2227	1461	1461	1461
Controls	✓	✓	✓	✓	✓	✓
Job Sector FE		✓	✓		✓	✓

Notes: This Table shows OLS estimates of (2). The dependent variable is the log of food consumption as a share of family income. Jobs Act is a dummy equal to 1 if the worker is hired after March 7, 2015. The controls are the same as in the previous table. The sample includes only large firm workers in 2018. Standard errors, clustered at the individual level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 5: Imputed Consumption

	Large Firms			Small Firms		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Log of Imputed Consumption						
Jobs Act	-8.58**	-8.68**	-7.91**	-1.65	0.44	-1.90
	(3.58)	(3.59)	(3.62)	(5.18)	(5.30)	(5.13)
Panel B: Consumption to Income Ratio						
Jobs Act	-6.64***	-7.09***	-6.64***	-3.85	-2.97	-2.80
	(2.45)	(2.46)	(2.46)	(3.52)	(3.64)	(3.63)
Observations	3753	3753	3753	1460	1460	1460
Sector FE		✓	✓		✓	✓
Risk Aversion			✓			✓

Notes: This Table shows the OLS estimates for equation (2). The dependent variable is the BPP-imputed log of consumption (panel A) and the imputed consumption-to-income ratio (panel B). Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 6: Small Firm Workers

	Food Share			Log Food Share		
	(1)	(2)	(3)	(4)	(5)	(6)
Jobs Act	-0.24	-0.14	-0.18	0.30	0.83	0.56
	(0.77)	(0.80)	(0.79)	(2.91)	(3.01)	(2.98)
Family Net Income (€100)			-0.22***			
			(0.04)			
Household Income (Log x 100)						-0.18***
						(0.03)
Observations	1428	1428	1428	1428	1428	1428
Controls	✓	✓	✓	✓	✓	✓
Job Sector FE		✓	✓		✓	✓

Notes: This Table shows OLS estimates of (2). The dependent variable is either food consumption as a share of family income (first three columns) or its log (last three columns). Jobs Act is a dummy equal to 1 if the worker is hired after March 7, 2015. The controls are the same as in the previous table. The sample includes only small firm workers in 2018. Standard errors, clustered at the individual level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 7: Public Sector Workers

	Food Share			Log Food Share		
	(1)	(2)	(3)	(4)	(5)	(6)
Jobs Act	0.15	0.10	0.31	1.45	1.16	1.54
	(0.87)	(0.87)	(0.86)	(3.25)	(3.23)	(3.18)
Family Net Income (€100)			-0.23***			
			(0.02)			
Household Income (Log x 100)						-0.21***
						(0.02)
Observations	3632	3632	3632	3632	3632	3632
Controls	✓	✓	✓	✓	✓	✓
Job Sector FE		✓	✓		✓	✓

Notes: This Table shows OLS estimates of (2). The dependent variable is either food consumption as a share of family income (first three columns) or its log (last three columns). Jobs Act is a dummy equal to 1 if the worker is hired after March 7, 2015. The controls are the same as in the previous table. The sample includes only public-sector workers in 2018. Standard errors, clustered at the individual level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 8: Bigger Firms ($\geq$ 20 employees)

	Food Share			Log Food Share		
	(1)	(2)	(3)	(4)	(5)	(6)
Jobs Act	-1.58**	-1.70**	-1.66**	-6.30**	-6.64**	-6.35**
	(0.74)	(0.75)	(0.74)	(3.14)	(3.16)	(3.15)
Family Net Income (€100)			-0.22***			
			(0.03)			
Household Income (Log x 100)						-0.20***
						(0.03)
Observations	2041	2041	2041	2041	2041	2041
Controls	✓	✓	✓	✓	✓	✓
Job Sector FE		✓	✓		✓	✓

Notes: This Table shows OLS estimates of (2). The dependent variable is either food consumption as a share of family income (first three columns) or its log (last three columns). Jobs Act is a dummy equal to 1 if the worker is hired after March 7, 2015. The controls are the same as in the previous table. The sample consists of workers from firms with at least 20 employees in northern Italy. Standard errors, clustered at the individual level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 9: Controlling for Risk Aversion and Tenure

	Food Share			Log Food Share		
	(1)	(2)	(3)	(4)	(5)	(6)
Jobs Act	-1.51**	-2.61**	-2.83**	-5.79**	-10.12**	-10.72**
	(0.69)	(1.24)	(1.26)	(2.88)	(5.02)	(5.08)
Family Net Income (€100)	-0.23***	-0.23***	-0.11**			
	(0.03)	(0.03)	(0.05)			
Household Income (Log x 100)				-0.21***	-0.21***	-0.12**
				(0.03)	(0.03)	(0.06)
Observations	2227	2227	435	2227	2227	435
Controls	✓	✓	✓	✓	✓	✓
Preference FE	✓	✓	✓	✓	✓	✓
Tenure FE		✓	≤ 5 years		✓	≤ 5 years

Notes: This Table shows OLS estimates of (4). The dependent variable is either food consumption as a share of family income (first three columns) or its log (last three columns). Jobs Act is a dummy equal to 1 if the worker is hired after March 7, 2015. The controls are the same as in the previous table. The sample consists of workers from firms with at least 20 employees in northern Italy. Standard errors, clustered at the individual level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 10: Heterogeneity Analysis

	By Age		By Education		By Income		Union Representation	
	< 40 y.o.	≥ 40 y.o.	No College	College	Below Median	Above Median	No	Yes
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Food Share								
Jobs Act	-1.94***	-1.35	-1.57**	-2.22***	-2.28***	-3.14***	-1.60*	-3.41***
	(0.72)	(0.99)	(0.76)	(0.82)	(0.75)	(0.84)	(0.83)	(0.78)
Panel B: Log of Food Share								
Jobs Act	-10.22***	-5.24	-6.11**	-11.51***	-9.49***	-15.11***	-7.38**	-15.27***
	(2.93)	(3.72)	(2.84)	(3.48)	(2.83)	(3.55)	(3.24)	(3.17)
Observations	1304	2385	2562	1126	1872	1818	1244	2445

Notes: This Table shows OLS estimates of (2). The dependent variable is either food consumption as a share of family income (Panel A) or its log (Panel B). Jobs Act is a dummy equal to 1 if the worker is hired after March 7, 2015. The sample is split according to the column definitions. The whole sample includes people from the North and the South of Italy. Standard errors, clustered at the individual level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 11: Assigned Parameters

Parameter	Description	Value	Source
ρ	Persistence of AR(1)	0.95	Jappelli et al. (2024)
σ_ε	Std of the shock	0.50	Jappelli et al. (2024)
w	Wage Rate	1	Normalization
b_0	UI Replacement Rate	0.75	Italian UI
b_1	""	0.54	""
r	Annual Interest Rate	0.02	Jappelli et al. (2024)

Table 12: MSM Performance and Estimated Parameters

Parameter	Description	Value	Std. Error
γ	Relative Risk Aversion	1.50287	0.08980
β	Discount Factor	0.96966	0.00090
s_{COL}	Separation Rate for COLs	0.04740	0.00118
s_{JA}	Separation Rate for JAs	0.12882	0.00760
f	Job Finding Rate	0.44290	0.01927

Moment	Source	Model	Data
Consumption Difference	Empirical Estimate	-7.89	-7.91
Mean Wealth-to-Income Ratio	SHIW 2016	6.64	6.18
Median Wealth-to-Income Ratio		4.33	4.37
P90 Wealth-to-Income Ratio		13.38	13.74
Mean Years of Tenure	LFS 2018	18.95	18.30
Unemployment Rate	EUROSTAT 2016-2019	11.83	11.09
Long Run Unemployment		6.57	6.80

Table 13: Regression with Simulated Data

Dependent Variable:	Log of Consumption × 100			Wealth / Income	
	(1)	(2)	(3)	(4)	(5)
Jobs Act	-7.890***	-2.741***	-5.303***	-0.660***	0.615***
	(0.115)	(0.104)	(0.378)	(0.029)	(0.025)
PDV of Income		1.798***	1.786***		0.387***
		(0.004)	(0.004)		(0.001)
Initial Wealth		0.725***	0.735***		0.567***
		(0.008)	(0.008)		(0.002)
Jobs Act x PDV of Income			0.095***		
			(0.013)		
Jobs Act x Init. Wealth			-0.077***		
			(0.022)		
Observations	1000000	1000000	1000000	1000000	1000000

Notes: This Table shows the OLS estimated coefficient of equation (5), run with simulated data. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 14: Optimal UI for JA and COL workers

Parameter	**Description**	**Value**
Optimal (type-variant) UI Structure		
$b_{0,COL}$	Proportional UI for COLs	0.67
$b_{1,JA}$	Cap of UI for COLs	0.51
$b_{0,COL}$	Proportional UI for COLs	0.83
$b_{1,JA}$	Cap of UI for JAs	0.63
Baseline (type-invariant) UI Structure		
b_0	Proportional Component	0.75
b_1	Cap Component	0.54

Notes: This the results of the maximization problem in (10)

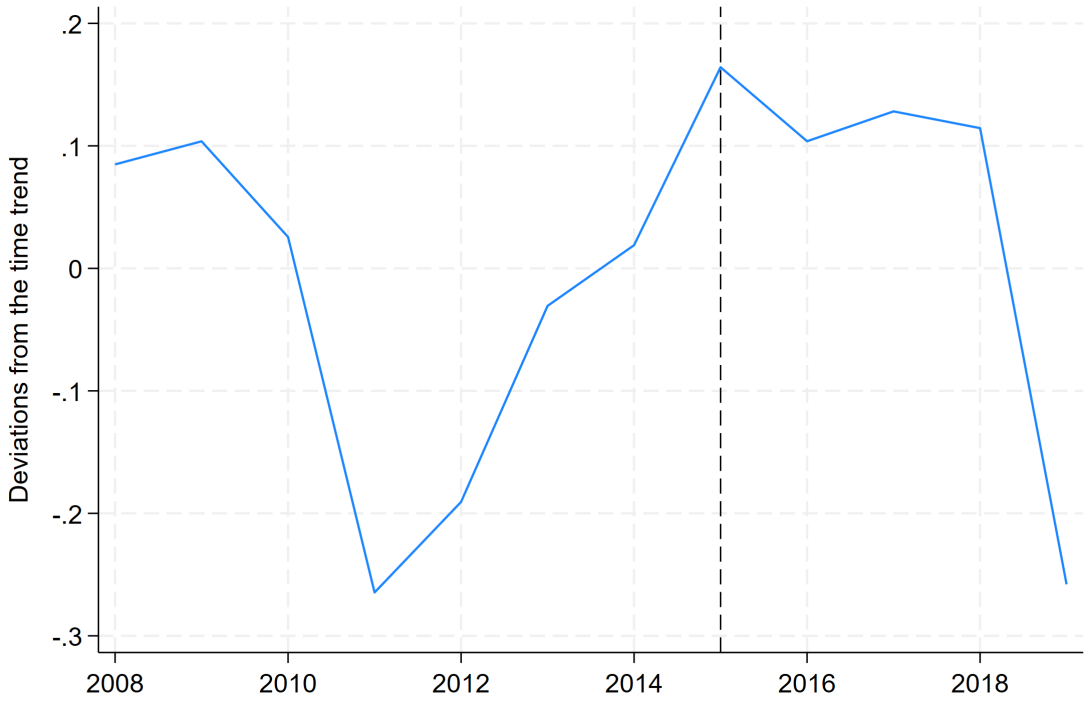
Table 15: Results with Optimal UI

Dependent Variable:	Log of Consumption × 100			Wealth / Income	
	(1)	(2)	(3)	(4)	(5)
Jobs Act	-6.257***	-2.110***	-4.078***	-1.400***	-0.214***
	(0.115)	(0.104)	(0.384)	(0.029)	(0.025)
PDV of Income		1.799***	1.790***		0.387***
		(0.004)	(0.004)		(0.001)
Initial Wealth		0.731***	0.738***		0.572***
		(0.008)	(0.008)		(0.002)
Jobs Act x PDV of Income			0.071***		
			(0.013)		
Jobs Act x Init. Wealth			-0.058***		
			(0.022)		
Observations	1000000	1000000	1000000	1000000	1000000

Notes: This Table shows the OLS estimated coefficient of equation (5), run with simulated data. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

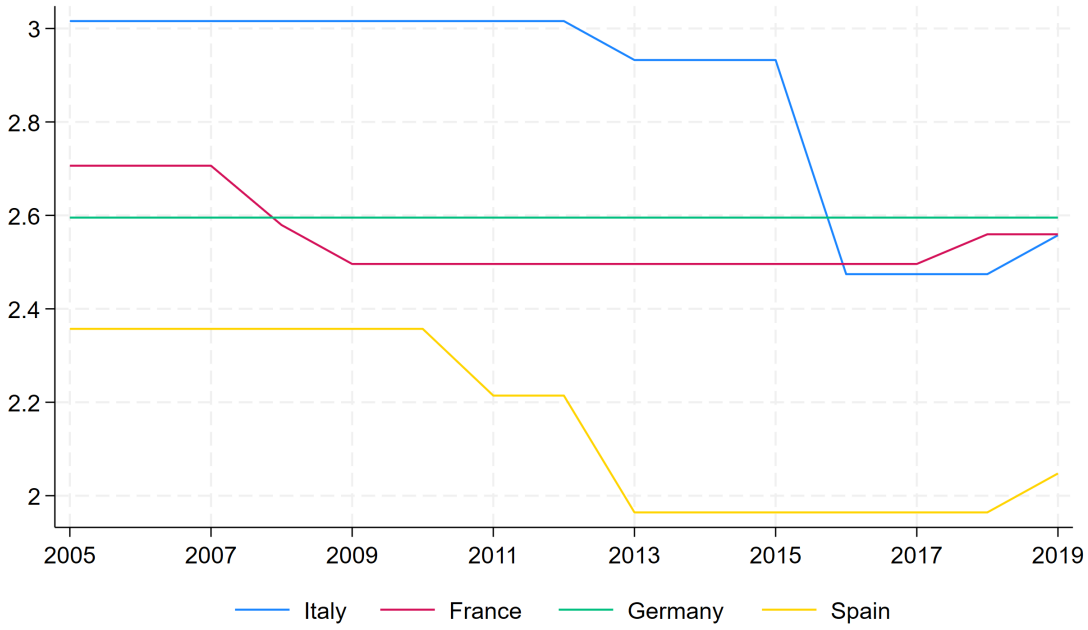
Figures

Figure 1: Workers' tenure in large firms over time.



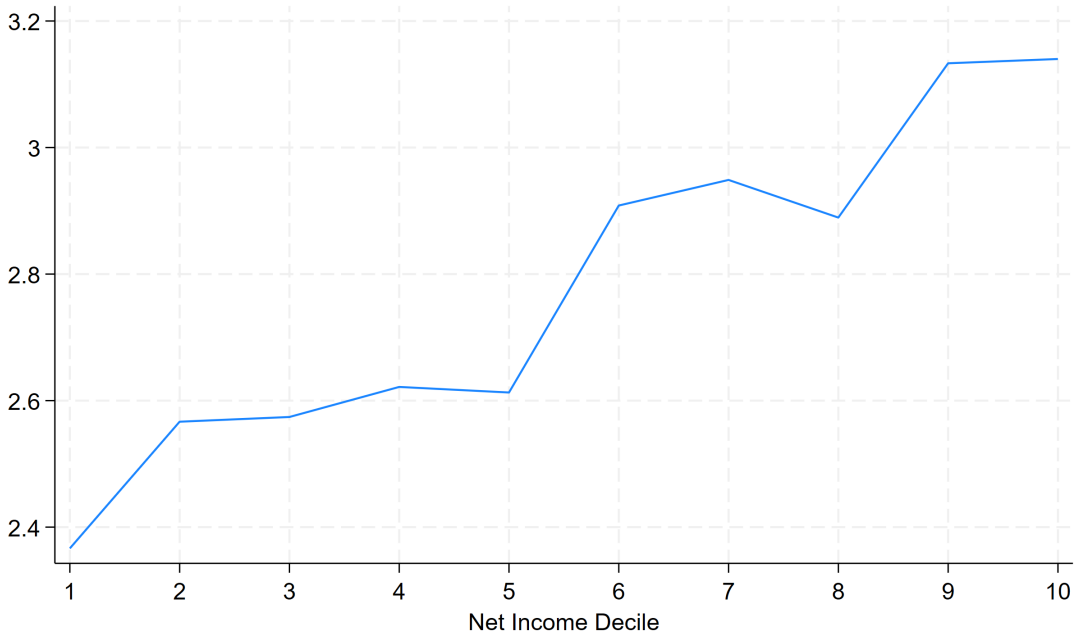
Notes: The Figure shows the average tenure of Italians employed in large firms between 2008 and 2019. Data are detrended to account for the business cycle (i.e., the 2008-2011 double-dip recessions). Average tenure increases until 2016 and then goes down sharply.

Figure 2: Employment Protection in OECD countries



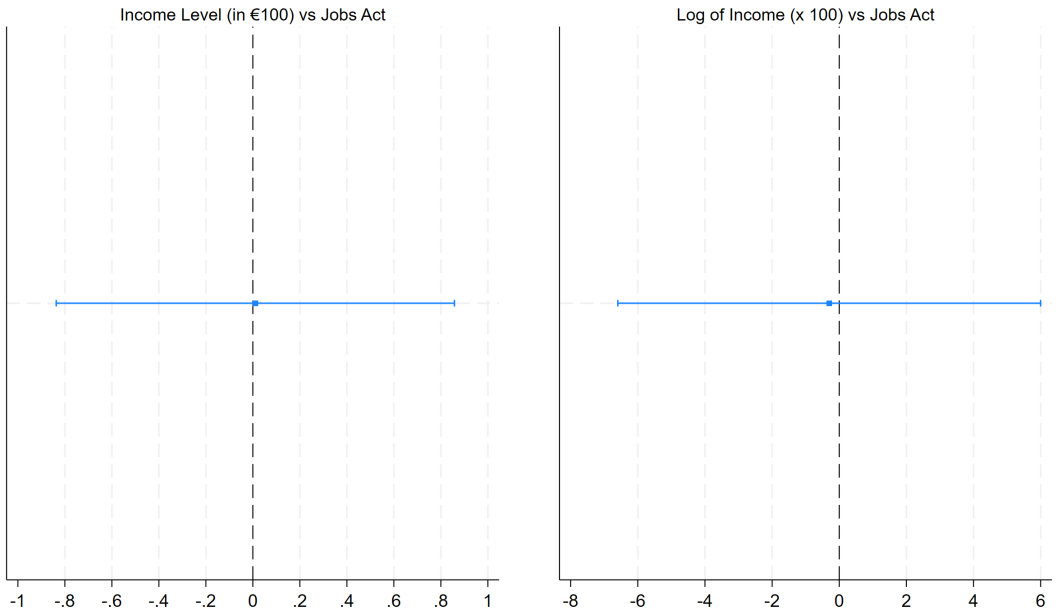
Notes: The Figure shows the employment protection index computed by the OECD. Italy shows a sharp drop in 2015 and 2016, represented by the blue line at the top. It's interesting to notice that even after the 2012 reform, Italy had a more rigid labor market than many of the OECD countries.

Figure 3: Ratio between Total and Food Consumption



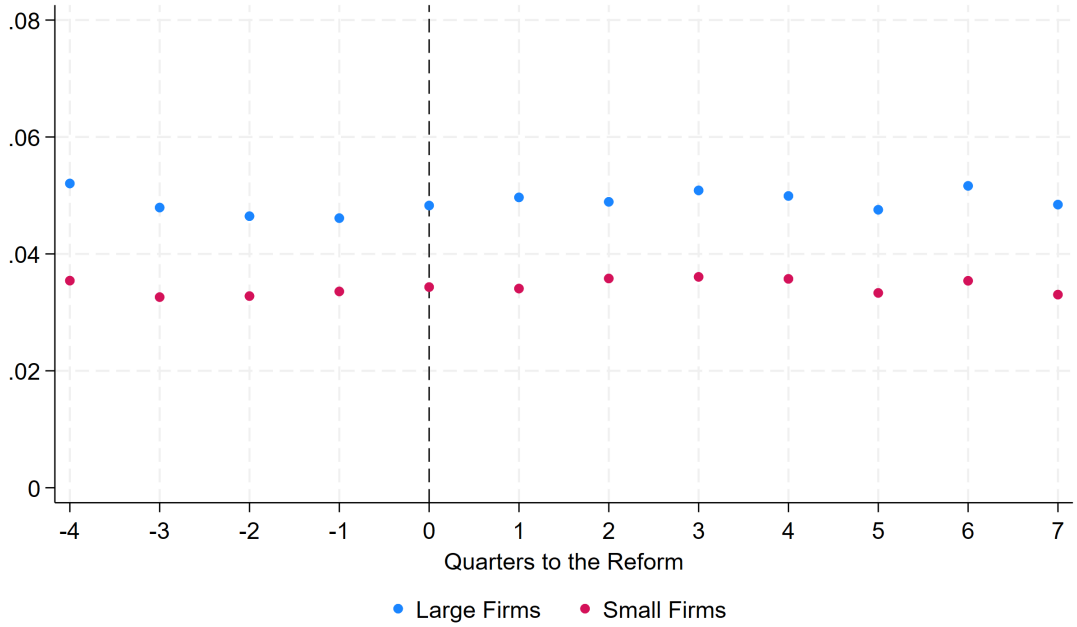
Notes: This graph shows the average ratio between total consumption and food consumption as a function of net income decile. This is obtained with the 2016 SHIW wave.

Figure 4: Income vs Jobs Act



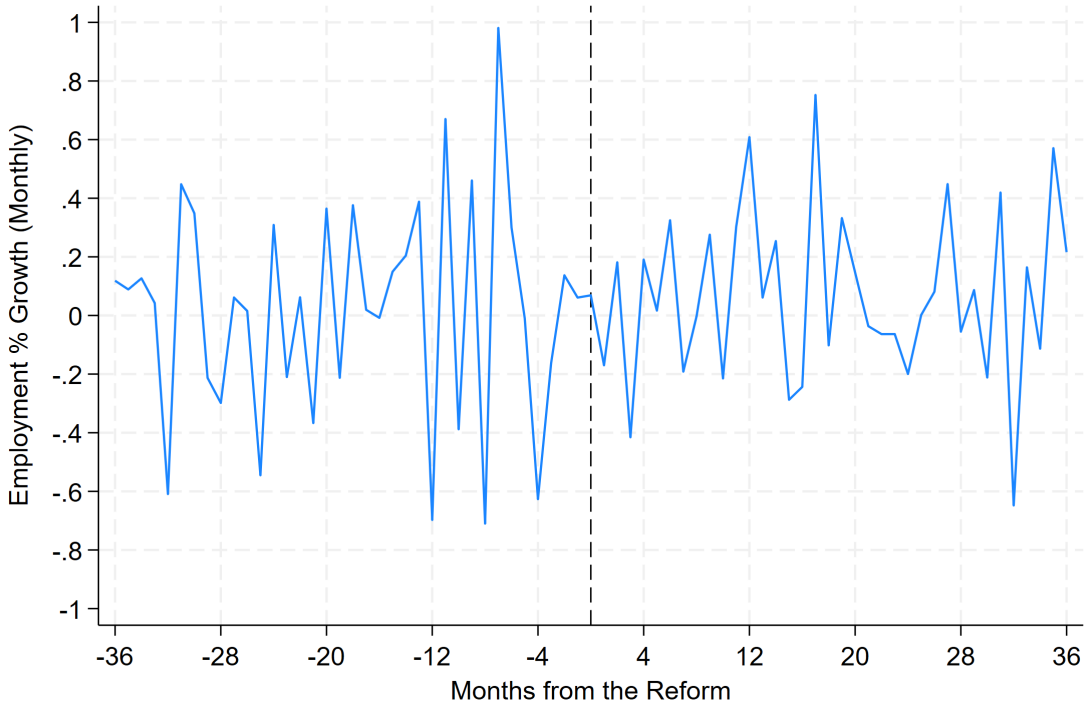
Notes: This Figure shows the OLS-estimated coefficient of income (both in levels and in logs) and the Jobs Act dummy, controlling for the same variables used in the main analysis.

Figure 5: Hiring Rates in Small and Large Firms



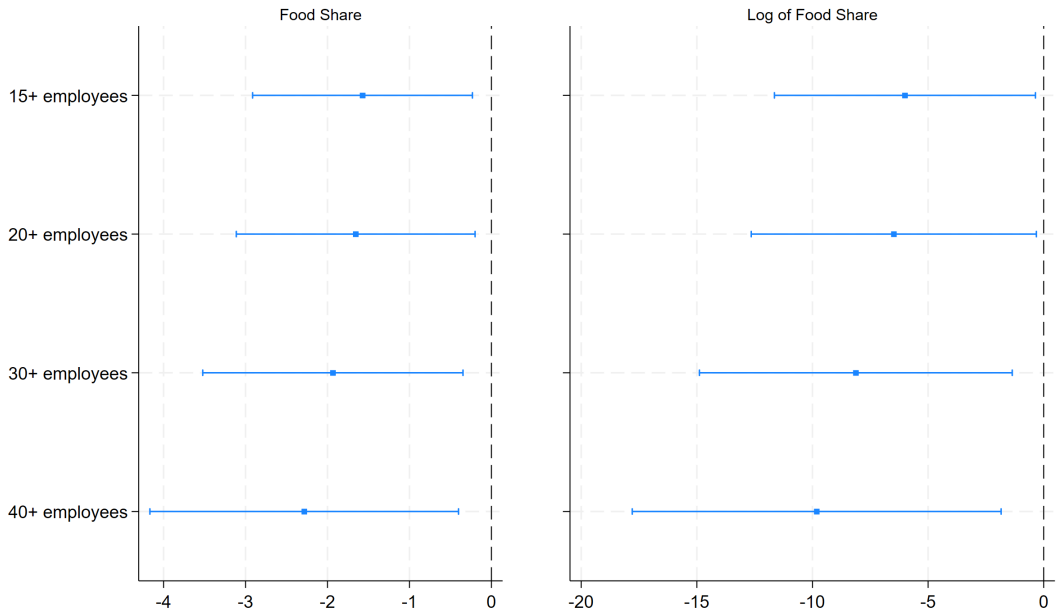
Notes: This Figure shows the sample hiring rates before and after the reform (first quarter 2015, t = 0). The horizontal axis shows quarters from 2014 (-4 is 2014-q1) to 2019 (7 is 2019-q4). The Source is the Italian Labor Survey.

Figure 6: Monthly Net Employment Growth



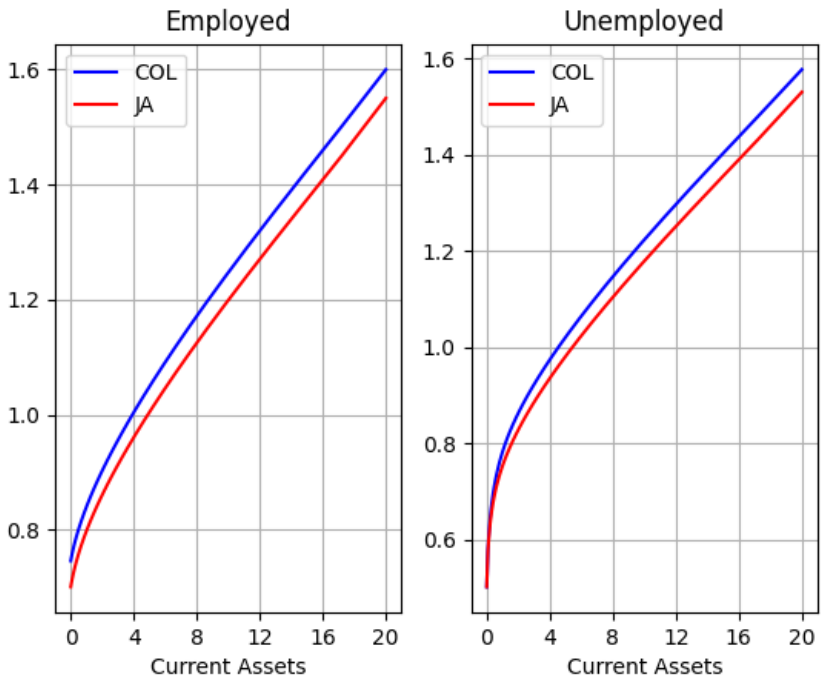
Notes: This Figure shows the monthly growth rate of employment. The Source is the Italian Statistical Office.

Figure 7: Estimated Effects for Bigger Firms



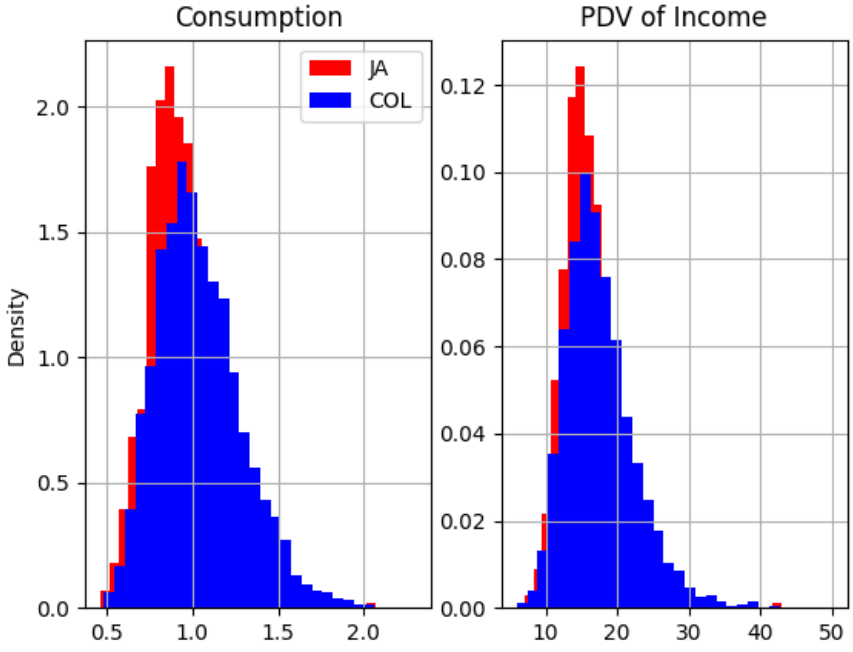
Notes: This Figure shows the effect on food consumption for employees of firms with +20, +30, and +40 workers.

Figure 8: Consumption Policy Functions by employment status



Notes: This Figure plots consumption as a function of current-period assets for JA and COL workers (red and light) in the median productivity state for employed and unemployed people (left and right panels, respectively).

Figure 9: Distribution of Consumption and PDV of Income



Notes: This Figure plots the distribution of consumption and of the PDV of income for JA and COL.

Appendix for:

Employment Protection and Consumption: Evidence from Italy

Nello Esposito

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Appendix A: Additional Survey Questions

Questions on the discount rate and the certainty equivalent.

Imagine receiving as a gift an amount of money equal to your annual net income. Would you give up this gift in exchange for the same amount next year, increased by...

1. *1 % of your annual income*
2. *5 % of your annual income*
3. *10 % of your annual income*
4. *50 % of your annual income*
5. *100 % of your annual income*
6. *300 % of your annual income*

.

Imagine you are being offered a lottery ticket for which you have a 50 % chance of winning your net annual income. Would you exchange the ticket for...

1. *5 % of your annual income*
2. *10 % of your annual income*
3. *25 % of your annual income*
4. *50 % of your annual income*
5. *75 % of your annual income*

.

The question on unforeseen spending is as follows.

Which amount of unforeseen spending would you be able to undertake without relying on external help from friends and/or relatives?

1. *Less than € 300*
2. *Between € 300 and € 800*
3. *Between € 800 and € 2000*
4. *More than € 2000*
5. *No expense*

.

Appendix B: Additional Tables and Figures

Table B1: What happens in case of unjust dismissal in JA vs COL

	Chart of Labour	**Jobs Act**
Who applies to:	Large Firms Workers	Large Firm Workers
Hiring Day:	Until March 6, 2015	March 7, onwards
Discriminatory Firing:	Reinstatement (i.e. Back in the Job)	Reinstatement
Disciplinary Firing:	Reinstatement	No Reinstatement
Other Terminations:	Probable Reinstatement	No Reinstatement
Monetary Insurance:	Arrears + Fine	Monetary Compensation (Function of Tenure)

Notes: This Table summarizes the main difference between *Jobs Act* and *Chart of Labour* workers.

Table B2: The Effect on Labor Supply

	(1)	(2)	(3)	(4)	(5)
Panel A: North of Italy					
Jobs Act	1.31***	1.10**	1.08**	0.09	-0.36
	(0.47)	(0.47)	(0.46)	(0.86)	(0.88)
Individual Net Income (€100)	0.62***	0.60***	0.60***	0.60***	0.64***
	(0.04)	(0.04)	(0.04)	(0.04)	(0.08)
Observations	2105	2105	2105	2105	414
Panel B: South and Centre					
Jobs Act	0.62	0.60	0.57	-0.78	-1.24
	(0.65)	(0.64)	(0.64)	(0.99)	(1.02)
Individual Net Income (€100)	0.69***	0.68***	0.67***	0.68***	0.83***
	(0.05)	(0.05)	(0.05)	(0.05)	(0.12)
Observations	2105	2105	2105	2105	414
Controls	✓	✓	✓	✓	✓
Job Sector FE		✓	✓	✓	✓
Risk Aversion			✓	✓	✓
Tenure				✓	≤ 5

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B3: Income Differences between JAs and COLs

	Income Level (€100)		Log of Income (× 100)	
	(1)	(2)	(3)	(4)
Jobs Act	-0.07	0.04	-0.84	-0.66
	(0.45)	(0.45)	(3.32)	(3.38)
Male	3.99***	3.53***	28.45***	25.76***
	(0.18)	(0.39)	(1.31)	(2.98)
Married	0.74***	1.27**	4.26***	6.23
	(0.21)	(0.55)	(1.57)	(4.11)
Family Size	-0.36***	-0.20	-2.80***	-1.74
	(0.09)	(0.17)	(0.68)	(1.29)
N. of Children	0.71***	0.86***	4.82***	5.85**
	(0.13)	(0.32)	(0.90)	(2.37)
Between 25 and 29	-1.51***	-0.71	-5.33**	0.14
	(0.36)	(0.76)	(2.65)	(5.53)
Between 30 and 39	-1.04***	-0.43	-3.29	0.69
	(0.27)	(0.72)	(2.00)	(5.33)
Between 40 and 49	-0.03	0.16	1.40	2.45
	(0.24)	(0.80)	(1.69)	(5.98)
Years of Education	0.63***	0.68***	4.10***	4.80***
	(0.03)	(0.07)	(0.20)	(0.49)
Observations	3505	703	3505	703
Job Sector	✓	✓	✓	✓
Tenure FE	All	≤ 5 years	All	≤ 5 years

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B4: Further Heterogeneity

	By Age		By Civil Status		By City Size		By Education	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Jobs Act	-2.94***	-12.75***	-2.89***	-11.81***	-3.14***	-12.26***	-2.36**	-8.70**
	(0.86)	(3.72)	(0.86)	(3.83)	(0.81)	(3.31)	(0.92)	(3.59)
Jobs Act x Above 45	0.72	4.42						
	(1.44)	(5.88)						
Above 45	2.26***	8.14***						
	(0.60)	(2.38)						
Jobs Act x Married			-0.25	-1.02				
			(1.34)	(5.61)				
Married			2.92***	11.12***				
			(0.57)	(2.24)				
Jobs Act x Big City					-1.41	-7.11		
					(1.39)	(6.04)		
Big City					-0.42	-1.03		
					(0.62)	(2.41)		
Jobs Act x College							-0.42	-2.90
							(1.32)	(5.61)
College							-4.24***	-18.07***
							(0.63)	(2.55)
Observations	2228	2228	2228	2228	2228	2228	2228	2228

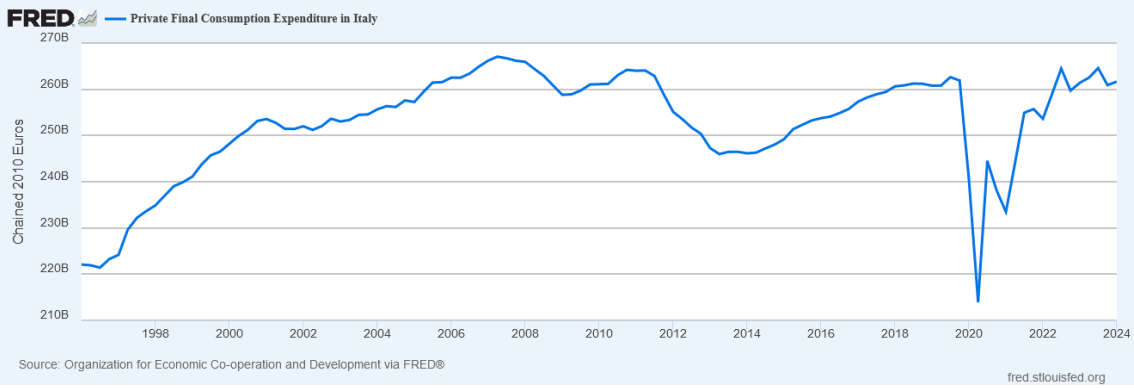
Notes: This Table show the estimates of (2) where the *Jobs Act* dummy is interacted with the heterogeneity dimension of interest. The dependent variable in odd columns is food consumption as a share of family income, and in even columns it is the log. The sample includes only large-firm workers in the North. Jobs Act is a dummy equal to 1 if the worker is hired after March 7, 2015. The controls are the same as in the previous table. The sample consists of workers from firms with at least 20 employees in northern Italy. Standard errors are clustered at the individual level and are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B5: HtM versus North

	All HtM			Poor HtM	Wealthy HtM
	(1)	(2)	(3)	(4)	(5)
North	-0.11***	-0.10***	-0.09***	-0.04***	-0.05***
	(0.02)	(0.02)	(0.02)	(0.01)	(0.01)
Male		0.01	0.01	-0.00	0.01
		(0.02)	(0.02)	(0.01)	(0.01)
Married		0.04**	0.03*	-0.02	0.05***
		(0.02)	(0.02)	(0.02)	(0.02)
College		-0.07***	-0.06***	-0.02*	-0.04**
		(0.02)	(0.02)	(0.01)	(0.02)
Net Wealth		-0.00***	-0.00***	-0.00***	0.00
		(0.00)	(0.00)	(0.00)	(0.00)
Familiy Labour Earnings		-0.00***	-0.00***	-0.00***	-0.00
		(0.00)	(0.00)	(0.00)	(0.00)
Observations	2148	2148	2148	2148	2148
Sector FE	·	·	✓	✓	✓

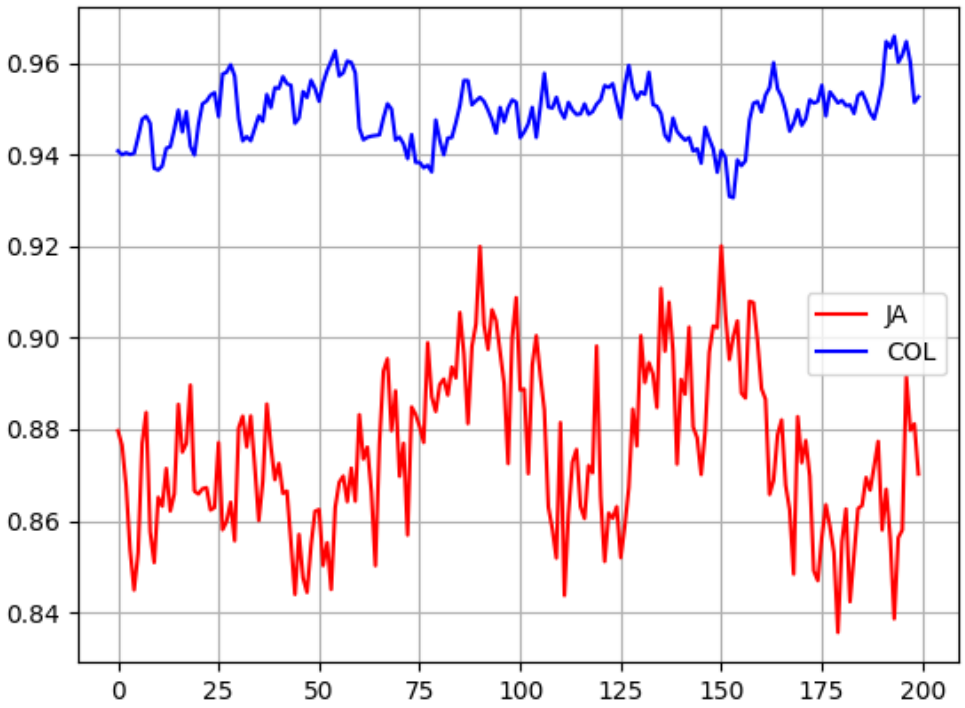
Notes: This Table shows the estimated coefficients of a regression of the HtM dummy against controls. The HtM dummy equals 1 for households whose liquid wealth is less than half of monthly income. Wealthy HtMs are those who, in addition to having liquid wealth below half their monthly income, are above the 25th percentile of the illiquid wealth distribution. Poor HtMs are the remaining HtMs who are not wealthy. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Figure B1: Aggregate Private Final Expenditure in Italy



Notes: This picture shows the time series of aggregate private consumption at the quarterly level. Aggregate consumption in Q4 2024 was still below the peak reached in Q2 2007, a quarter before the Great Financial Crisis. *Source:* St. Louis Federal Reserve.

Figure B2: Simulated Income Paths for JA and COL workers



Notes: This picture shows the simulated incomes of JA and COL workers over time in the model.

Appendix C: Blundell-Pistaferri-Preston Imputation

In their seminal work, Blundell et al. (2008) impute nondurable consumption in the PSID from CEX. Their methodology consists of estimating a demand equation of food consumption - available in both surveys - using non-food consumption, demographic characteristics, and prices as RHS variables. Under some conditions, this demand function can be inverted, and non-food consumption might be pinned down by applying the estimated coefficients to the variables in PSID. A recent work by Patterson (2023) uses a similar approach.

In this Appendix, I impute nondurable consumption into PLUS by estimating a demand function using HBS. In practice, HBS is the counterpart of CEX in BPP, and PLUS is the counterpart of PSID in BPP. Table D1 shows the summary statistics of HBS, while Table D2 the results of the imputation (OLS estimation results of (3). Finally, Figure C1 compares the distribution of the actual log of spending (right panel) with the imputed (left panel).

Table C1: Summary Statistics of HBS

	Mean	Sd
Age 18-34	0.12	0.32
Age 35-64	0.88	0.32
Male	0.68	0.47
Married or Civil Union	0.54	0.50
South	0.29	0.45
Centre	0.21	0.41
North	0.50	0.50
Total spending (monthly)	2625.06	1346.57
Food-only spending (monthly)	470.06	274.42
Total/Food Consumption Ratio	6.75	4.08

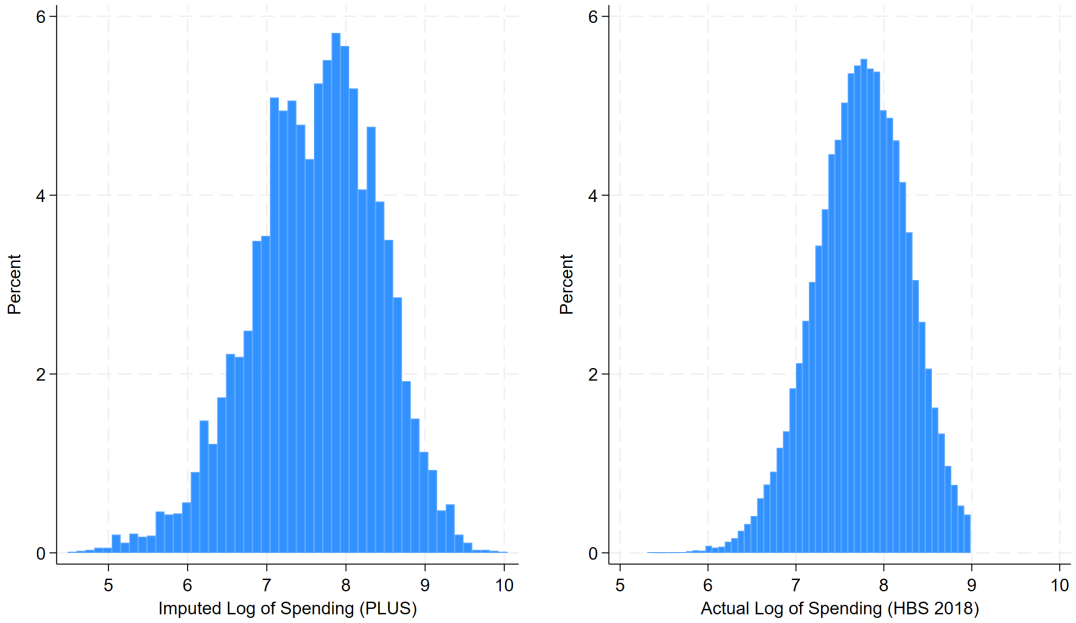
Table C2: Results of the Imputation

	(1)
Log of Total Spending	0.70***
	(0.01)
Family size	0.10***
	(0.00)
Male	-0.00
	(0.00)
Age 35-64	0.06***
	(0.01)
Married or Civil Union	0.06***
	(0.01)
Italian citizenship	-0.07***
	(0.01)
Centre	-0.11***
	(0.01)
North	-0.19***
	(0.00)
Middle education	-0.07***
	(0.01)
High education	-0.18***
	(0.01)
Constant	0.48***
	(0.04)
Observations	52774
R^2	0.472

Standard errors in parentheses

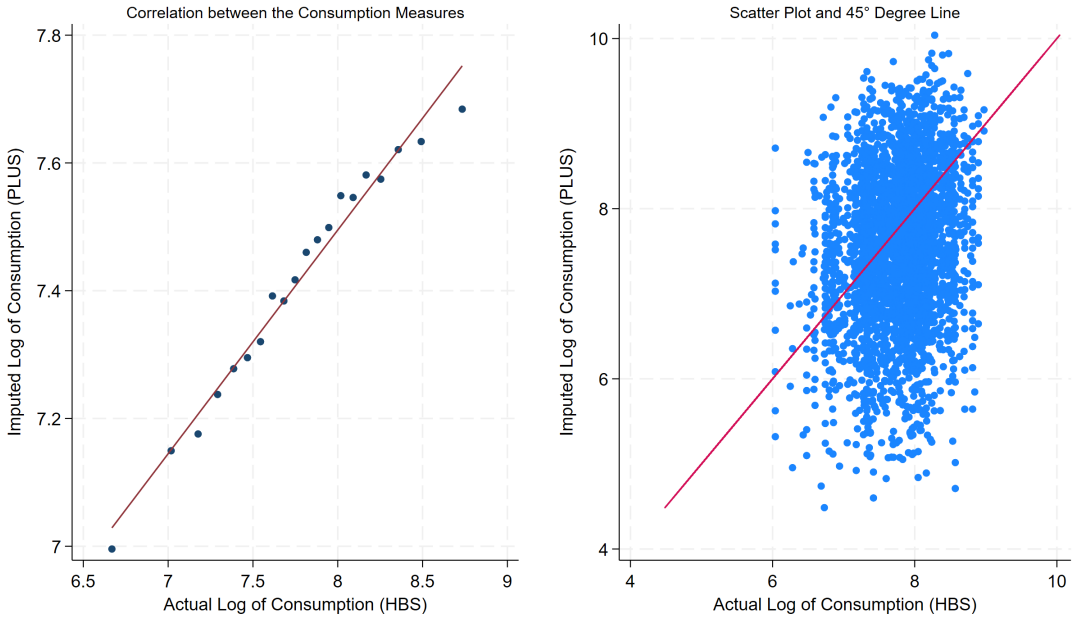
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Figure C1: Distribution of Imputed and Actual Consumption



Notes: This Figure shows the pooled distribution of the imputed measure of consumption, obtained using the BPP procedure.

Figure C2: Percentiles and Scatter Plot of the Distributions



Notes: The left panel shows the correlation between the percentiles of the actual distribution and the percentiles of the imputed distribution. The right graph shows the scatter plots with a 45-degree line.

Appendix D: Differences in Difference

I estimate the following equation via OLS:

$$Y_{it} = \beta Jobs\ Act_{it} \times Large_{it} + \delta Jobs\ Act_{it} + \gamma Large_{it} + \varphi \mathbf{X}_{it} + \lambda_t + \varepsilon_{it}$$

Where Y_{it} is the food share, the log of food spending, and weekly hours worked. The following Table shows the estimation results. The first two columns show results for the food share, the middle ones the log of food spending, and the last one the hours worked. Within each dependent variable, I perform two specifications: without and with controls. To limit selection problems as much as possible, I drop workers from firms with between 10 and 14 employees.

Table D1: Difference-in-Differences Estimates

	Food Share	Hours Worked	Imputed Consumption
	(1)	(2)	(3)
Jobs Act x Large	-1.41*	1.05	-12.03**
	(0.83)	(0.71)	(6.08)
Employed in a Big Firm	-1.20***	3.26***	12.53***
	(0.26)	(0.21)	(1.85)
Jobs Act	-0.89	0.24	-0.03
	(0.69)	(0.62)	(5.03)
Observations	12116	12608	11105

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Appendix E: Detailed Institutional Framework

The Italian labor market has historically been characterized by a certain degree of rigidity. In 1970, the COL was passed. The COL contains general rules on workplace safety, unionization, and, most importantly for the paper's argument, workers' reinstatement in cases of unjust dismissal by the employer. Article 18, in its first version, stated that employees of large firms got reinstated in their jobs in all cases of unjust dismissal, ruled by a labor judge. This means workers fired unlawfully were entitled to their jobs back, plus all forgone wages during their absence and damages from the employer.

This labor regulation made firing very hard and costly for employers. Indeed, judges in Italy have a strong pro-labor attitude (Del Punta 2025), and the overwhelming majority of labor cases are ruled in workers' favor. It also created a dual labor market: on one hand, a category of over-privileged workers for whom being fired was virtually impossible; on the other hand, other workers with much lower employment protection (and therefore higher employment risk).

Article 18 was rewritten in 2012. The new version of Article 18 covers all large firm employees hired until March 6, 2015. After the reform, which was named after Labor Minister Elsa Fornero[19], compulsory reinstatement remained only in some cases of unjust dismissal, while it was abolished in others. In those cases, the employer only has to pay the worker a fine but is no longer obliged to reinstate her. This new version of Article 18 gives judges broad discretion: they must not only decide whether a firing is unlawful, but also determine the reason for its unlawfulness. Given a pro-labor attitude, they typically find unjust dismissals based on circumstances that would have made reinstatement compulsory, since workers' preferred option in most cases is to get their jobs back. Therefore, employment protection remained high compared with other EU countries. This is evident in Figure 2: the drop in the employment protection index in 2012 is modest compared with the drop in 2015/2016 due to the *Jobs Act*.

Given that the 2012 reform failed to increase flexibility in the labor market, considered in the mid-2010s an economic policy priority in Italy, the Government led by Prime Minister Matteo Renzi passed the *Jobs Act*, applied to large-firm employees starting on March 7th 2015. Article 18 still applies to workers hired before this date, still by large firms. The major novelty is that, in case of unjust dismissals, workers get reinstated in their job only in case of discriminatory firing, i.e., in a circumstance in which a worker is fired because of her gender, race, sexual orientation, or political and religious beliefs. In all other cases of unjust dismissal ruled by the judge, the employer is not forced to reinstate or hire the worker back, but only to pay a fine proportional to her tenure with the firm. This intervention had two strong effects. First, it eliminated or significantly reduced judges' discretionary power, since, in almost all circumstances, the employer no longer has to reinstate the worker. Second, it reduced, or made much more predictable, firing costs for firms. Indeed, firms know exactly the cost they would incur in case of a "negative" ruling, since the amount payable to workers can be computed with certainty using

[19]This reform shall not be confused with the Fornero pension reform, another reform Minister Fornero is famous for.

the formula provided by the law. On the other hand, this reduced employment protection. The *Jobs Act* is only part of one of the packages of reforms undertaken by PM Renzi's Government to reform Italy's labor market. In December 2014, the Italian Parliament passed a law that allowed the Government to reform the labor market without further future approval by the Parliament (this kind of law is called *legge delega*, and it's sometimes justified by the complexity of the reforms, which is therefore best dealt with by experts from the Government). Other reforms addressed the regulation of mass layoffs, workplace safety and privacy, and unemployment insurance. Two concerns arise from this point. The first is the relation between the JA and these other reforms, and the second is whether both firms and workers expected such a reform. Regarding the former, these other pieces of legislation involve all workers regardless of firm size and the date they were hired. They are therefore uncorrelated with the JA and might affect consumption only through a fixed effect. Regarding the latter, the passing of the *legge delega* created expectations of a reform of Article 18. However, the Government intentionally kept the exact content of the reform secret and released it on March 6, 2015. Therefore, there was no scope for manipulation around March 7, 2015, either from the workers' side or the firm's side.

Appendix F: Model Appendix

F1 Recursive Formulation and Numerical Details

There are two types of consumers: JA and COL. They are homogeneous in their preference parameters and productivity but differ in employment risk: the former have a higher job separation rate s_j (with $j \in \{ JA, COL\}$ and $s_{JA} > s_{COL}$). The job finding rate f is the same for both. For both JAs and COLs, the Bellman Equations for the employed and unemployed workers are the following:

$$V_{jt}^{empl}(a,p) = \max_{c_j, a'_j} u(c_j) + \beta \left\{ s_j \mathbb{E}\left[V_{j,t+1}^{unem}(a'_j, p')|p\right] + (1-s_j)\mathbb{E}\left[V_{j,t+1}^{empl}(a'_j, p')|p\right] \right\}$$

$$a'_j + c_j = (1+r)a_j + w \times p_j, \quad a' \geq 0$$

$$V_{jt}^{unem}(a,p) = \max_{c_j, a'_j} u(c) + \beta \left\{ (1-f)\mathbb{E}\left[V_{jt}^{unem}(a'_j, p')|p\right] + f\mathbb{E}\left[V_{jt+1}^{empl}(a'_j, p')|p\right] \right\}$$

$$a'_j + c_j = (1+r)a_j + b(p_j), \quad a' \geq 0$$

The FOCs of the two Bellman equations are:

$$u'(c_{jt}^{empl}) \geq \beta \left\{ s_j \mathbb{E}\left[V_{ajt+1}^{unem}(a'_j, p')|p\right] + (1-s_j)\mathbb{E}\left[V_{ajt+1}^{empl}(a'_j, p')|p\right] \right\}$$

$$u'(c_{jt}^{unem}) \geq \beta \left\{ (1-f)\mathbb{E}\left[V_{ajt+1}^{unem}(a'_j, p')|p\right] + f\mathbb{E}\left[V_{ajt+1}^{empl}(a'_j, p')|p\right] \right\}$$

The Envelope Conditions are:

$$V_{ajt}^{empl}(a_j, p) = (1+r)u'(c_{jt}^{empl})$$

$$V_{ajt}^{unem}(a_j, p) = (1+r)u'(c_{jt}^{unem})$$

Where $j \in \{JA, COL\}$ and V_{aj} denotes the derivative of the value function with respect to current wealth for type j. I find the consumption policy functions for JA and COL workers - in employment and unemployment - via the endogenous grid point method (see Carroll 2006). After finding the policy functions, using the transition matrices for employment Π_{JA} and Π_{COL}, I simulate a series of 0 (if unemployed) and 1 (if employed) for N_{JA} and N_{COL} households and 200 periods. Then, I compute the mean difference between the log of consumption of JA and COL workers. Finally, I run standard OLS on the simulation outcome, controlling for wealth and the present discounted value of income, computed as the discounted sum of all income earned by the individual in the model, using the interest rate r for discounting.

F2 Epstein-Zin Preferences

The baseline model assumes isoelastic preferences, in which the *EIS* and relative risk aversion are reciprocals. In this appendix, I assume recursive preferences in the fashion of Epstein and Zin (1989), in which intertemporal substitution and risk aversion are governed by two separate parameters: the *EIS* θ and the risk aversion γ. The model with recursive preferences is the following:

$$\max_{c_j, a_{jt+1}} \quad V_{jt} = \left[(1-\beta) c_{jt}^{\frac{\theta-1}{\theta}} + \beta \mathbb{E}_t \left[V_{jt+1}^{1-\gamma} \right]^{\frac{\theta-1}{\theta(1-\gamma)}} \right]^{\frac{\theta}{\theta-1}}$$

The expectation is taken over the employment and productivity states: (e, p). Subject to the same constraints as in Section 7. The Table below shows the calibration, while the next table shows the estimated coefficients of equations (5) and (6).

Table F1: Calibration - EZ Preferences

Parameter	Value	Std Error	Moment	Model	Data
γ	1.74	0.146	Log Diff. in Consumption	-8.05	-7.91
eis	0.29	0.0054	Mean Wealth to Income	6.26	6.18
β	0.95	0.0051	Median Wealth to Income	4.61	4.37
	-	-	P90 Wealth to Income	13.42	13.74
s_{COL}	0.05	0.0051	COLs Years of Tenure	18.29	18.30
s_{JA}	0.12	0.0034	Unemployment Rate	11.53	11.09
f	0.43	0.115	Long Run Unemployment Rate	6.56	6.80

Table F2: Results - EZ Preferences

Dependent Variable:	Log of Consumption × 100			Wealth / Income	
	(1)	(2)	(3)	(4)	(5)
Jobs Act	-8.055***	-3.426***	-6.427***	-0.052***	0.706***
	(0.116)	(0.108)	(0.388)	(0.020)	(0.018)
PDV of Income		1.562***	1.549***		0.215***
		(0.004)	(0.005)		(0.001)
Initial Wealth		0.676***	0.678***		0.601***
		(0.011)	(0.012)		(0.002)
Jobs Act x PDV of Income			0.103***		
			(0.013)		
Jobs Act x Init. Wealth			-0.023		
			(0.030)		
Observations	1000000	1000000	1000000	1000000	1000000
COLs Mean				4.584	4.584

F3 Two-account model

Following Auclert et al. (2024) and Graves (2025), I also extend the model to allow consumers to hold two accounts: liquid and illiquid. The illiquid account yields a higher return, r_a, than the liquid account, r_b, but each consumer has an i.i.d. probability χ with which they can tap in or tap out of the illiquid account. When they adjust their illiquid account, consumers also pay a transaction cost equal to k percentage points of the absolute difference between the wealth they decide to hold next period and their current wealth. The rest of the model is the same as before. The recursive formulation follows below. The utility function is CRRA with risk aversion γ. In the Bellman equations, a and b denote the holdings of the illiquid and liquid accounts, respectively. In this model, there are 5 states: i) whether the consumer can adjust or not, ii) the holdings of the illiquid account, iii) the holdings of the liquid account, iv) employment status, and v) idiosyncratic productivity. I treat the first one as a state just for numerical convenience, but since χ is iid, it can be disregarded as a state. When the individuals can adjust, they face the following Bellman equation, with the respective budget constraints:

$$V_{jt}^{empl,adj}(a_j,b_j,p) = \max_{c_j,a'_j,b'_j} +\beta\chi\left\{s_j\mathbb{E}\left[V_{jt+1}^{unem,adj}(a'_j,b'_j,p')|p\right] + (1-s_j)\mathbb{E}\left[V_{jt+1}^{empl,adj}(a'_j,b'_j,p')|p\right]\right\} +$$

$$\beta(1-\chi)\left\{s_j\mathbb{E}\left[V_{jt+1}^{unem,noadj}(a'_j,b'_j,p')|p\right] + (1-s_j)\mathbb{E}\left[V_{jt+1}^{empl,noadj}(a'_j,b'_j,p')|p\right]\right\}$$

$$a'_j + b'_j = (1+r_a)a_j + (1+r_b)b_j + y(empl,p), \quad a'_j \geq 0, \quad b'_j \geq 0$$

$$V_{jt}^{unem,adj}(a_j,b_j,p) = \max_{c_j,a'_j,b'_j} +\beta\chi\left\{f\mathbb{E}\left[V_{jt+1}^{empl,adj}(a'_j,b'_j,p')|p\right] + (1-f)\mathbb{E}\left[V_{jt+1}^{unempl,adj}(a'_j,b'_j,p')|p\right]\right\} +$$

$$\beta(1-\chi)\left\{f\mathbb{E}\left[V_{jt+1}^{eml,noadj}(a'_j,b'_j,p')|p\right] + (1-f)\mathbb{E}\left[V_{jt+1}^{unem,noadj}(a'_j,b'_j,p')|p\right]\right\}$$

$$a'_j + b'_j = (1+r_a)a_j + (1+r_b)b_j + y(unem,p), \quad a'_j \geq 0, \quad b'_j \geq 0$$

When they cannot adjust, they face the following equations:

$$V_{jt}^{empl,noadj}(a_j,b_j,p) = \max_{c_j,a'_j,b'_j} +\beta\chi\left\{s_j\mathbb{E}\left[V_{jt+1}^{unem,adj}(a'_j,b'_j,p')|p\right] + (1-s_j)\mathbb{E}\left[V_{jt+1}^{empl,adj}(a'_j,b'_j,p')|p\right]\right\} +$$

$$\beta(1-\chi)\left\{s_j\mathbb{E}\left[V_{jt+1}^{unem,noadj}(a'_j,b'_j,p')|p\right] + (1-s_j)\mathbb{E}\left[V_{jt+1}^{empl,noadj}(a'_j,b'_j,p')|p\right]\right\}$$

$$c_j + b'_j = (1+r_b)b_j + y(empl,p), \quad a'_j = (1+r_a)a_j, \quad a'_j \geq 0, \quad b'_j \geq 0$$

$$V_{jt}^{unem,noadj}(a_j,b_j,p) = \max_{c_j,a'_j,b'_j} +\beta\chi\left\{f\mathbb{E}\left[V_{jt+1}^{empl,adj}(a'_j,b'_j,p')|p\right] + (1-f)\mathbb{E}\left[V_{jt+1}^{unempl,adj}(a'_j,b'_j,p')|p\right]\right\} +$$

$$\beta(1-\chi)\left\{f\mathbb{E}\left[V_{jt+1}^{eml,noadj}(a'_j,b'_j,p')|p\right] + (1-f)\mathbb{E}\left[V_{jt+1}^{unem,noadj}(a'_j,b'_j,p')|p\right]\right\}$$

$$c_j + b'_j = (1+r_b)b_j + y(empl,p), \quad a'_j = (1+r_a)a_j, \quad a'_j \geq 0, \quad b'_j \geq 0$$

Numerical Solution and Calibration. I obtain the model's policy functions via VFI. As before, I leverage the estimated difference in consumption between JAs and COLs to estimate γ. Then I estimate β and χ to match key moments of illiquid wealth to income and the self-reported MPC in SHIW 2016, following the calibration strategy in Auclert et al. (2024). I classify assets into liquid and illiquid accounts following Kaplan and Violante (2014) and Kaplan et al. (2018): the liquid account contains cash, government and corporate bonds, while the illiquid account contains real estate, stocks, managed funds, and foreign-denominated assets. The discount factor is 0.92. The adjustment probability is 0.10, meaning individuals adjust their illiquid account once every 10 years on average. The interest rate on the illiquid account is set at 10%, while the interest rate on the liquid account is -2%. Neither the liquid nor the illiquid account can be shorted. I calibrate the transition-matrix parameters in the same way as before. To reduce computation time, I set the grid size to 50 and the maximum grid value to 10. Table F3 shows the model calibration, while Table F4 shows differences in the wealth-to-income ratio for liquid, illiquid, and total accounts. JA workers have a higher wealth-to-income ratio overall, but liquid accounts mostly explain this difference; in other words, people prefer to save precautionarily in liquid wealth, a result consistent with Kaplan and Violante (2014). Figure F1 provides the graphical intuition. While the distributions of illiquid wealth for JA and COL almost overlap, the distribution of liquid wealth for JA almost first-order stochastically dominates that of COL. Finally, Figure F2 depicts the distributions of consumption and total wealth, i.e., the sum of the liquid and the illiquid accounts.

Table F3: Calibration of the Two-Account Model

Parameter	Value	Moment	Model	Data
γ	2.22	Log Diff. in Consumption	-7.89	-7.910
β	0.92	Average Illiquid Wealth to Income	4.11	4.12
χ	0.10	Average MPC	0.47	0.48
s_{COL}	0.046	COLs Years of Tenure	19.35	18.310
s_{JA}	0.132	Unemployment Rate	11.74	10.750
f	0.454	Long Run Unemployment Rate	6.33	6.500

Notes: The other parameters of the calibration, excluding the account grids, are the same as in the baseline model.

Table F4: Difference in the Wealth-to-Income ratios

%Δ Illiquid Account	%Δ Liquid Account	%Δ Total Wealth	%Δ Consumption
5.89	20.82	9.65	-7.89

Notes: These are the percentage differences of the wealth-to-income ratio.

Figure F1: Distribution of Illiquid and Liquid accounts

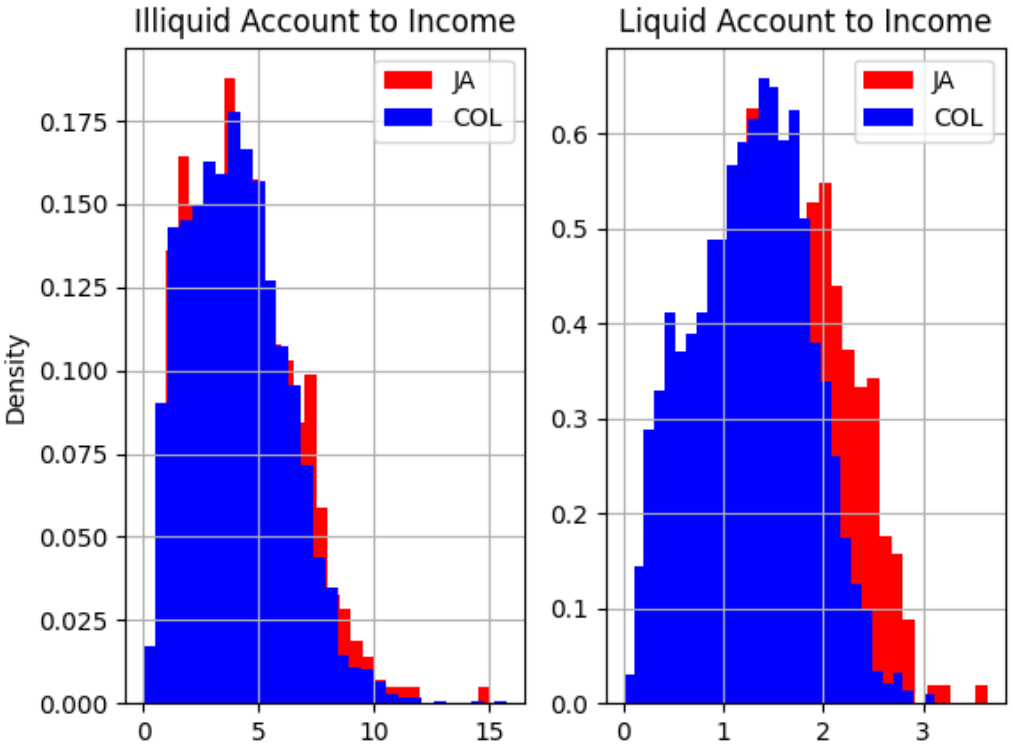
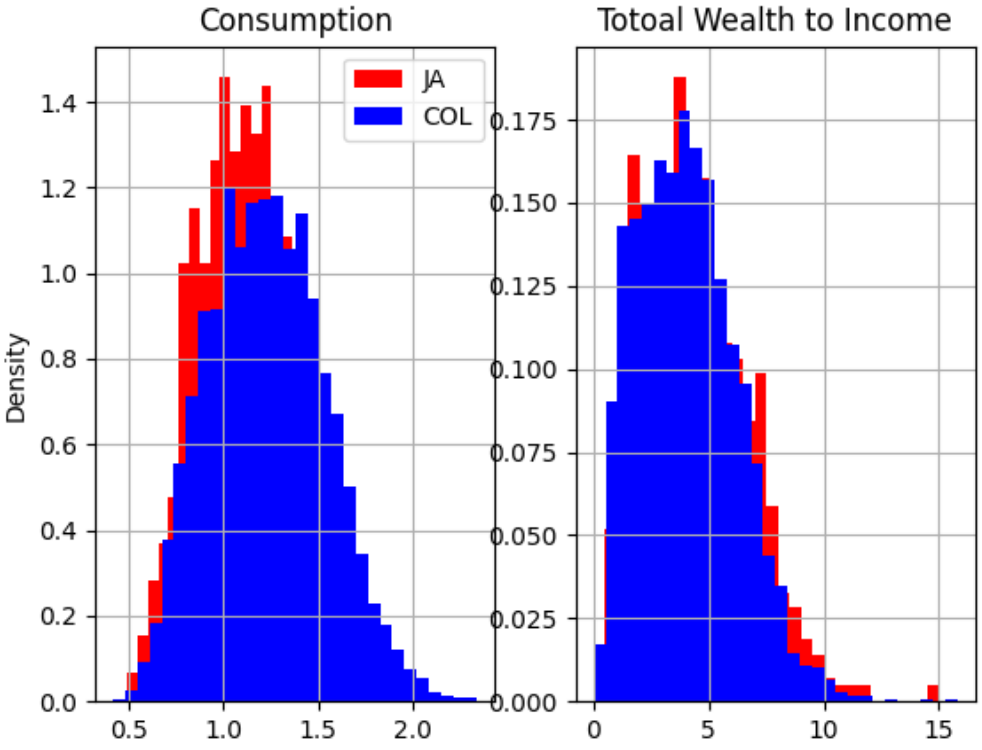


Figure F2: Total Wealth (Liquid + Illiquid)



F4 Life Cycle Version

Time is discrete. Workers enter the labor market at the age of 25, retire at 65, and die with certainty at the age of 90. In all periods of the model, mortality risk exists. The age-dependent survival probabilities (s_t) are sourced from Italy's demographic tables. Workers start their careers in the COL by assumption, and if they lose their jobs in subsequent periods of the model, the next jobs they take are subject to the JA. When people die, they leave a bequest, and the bequest motive is characterized by $B(a_{t+1})$. Workers face both employment risk and productivity risk, and their income is defined as before in both cases. Differently from the baseline, however, an age profile is attached to their income up to the moment they retire. When they retire, they get their last income multiplied by 0.75. This simplifying assumption rests on the fact that my model does not study the retirement period in detail, and that in the estimation I target only the wealth-to-income age profile during working years.

$$z_t = \begin{cases} y\big(e_t(e_{t-1}, \Pi_{jt}), p_t\big) \times \exp\{g(t)\} & \text{if} \quad t \leq T_{ret} - 1 \\ 0.75 \times y_{t-1} \exp\{g(T_{ret} - 1)\} & \text{if} \quad t > T_{ret} - 1 \end{cases} \tag{F1}$$

$$g(t) = a_0 + a_1 \times t + a_2 \times t^2 + a_3 \times t^3$$

The parameters of the AR(1) for the log of productivity are estimated via GMM using SHIW data from 2000 to 2016. The loadings on the age profile are the estimated coefficients from a regression of mean earnings on a cubic polynomial of age up to 64. The recursive formulation of the problem in the working period is the following:

$$V_t(X_t) = \max_{c_t, a_{t+1}} c_t^{1-\gamma}/(1-\gamma) + \beta \left\{ s_t \mathbb{E}_t \left[V_{t+1}(X_{t+1}) | X_t \right] + (1 - s_t) B(a_{t+1}) \right\}$$

$$a_{t+1} = (1 + r)a_t + z_t - c_t$$

Where $B(a_{t+1}) = a_{t+1}^{1-\gamma}/(1-\gamma)$ is the bequest function and X_t is the collection of the state variables: time, whether a worker is JA or COL, whether she is employed or unemployed, her productivity, and wealth.

In this version, the purpose of the MSM is to estimate the preference parameters γ and β. On top of the empirical result of -7.91%, I target the working-life profile of the wealth-to-income ratio. I obtain this last one using SHIW panel data from 2000 to 2016. I address the well-known multicollinearity problem among age, time, and cohort effects following Deaton and Paxson (1994): I regress the wealth-to-income ratio on age and cohort and year dummies, assuming the latter sum to 0 and are orthogonal to a time trend. I then compute the mean predicted life cycle profile for the following age ranges: 25-29, 30-34, 35-39, 40-44, 45-49, 50-54, 55-59, 60-64. Table F5 and Figure F3 show the model's overall performance and parameter estimates.

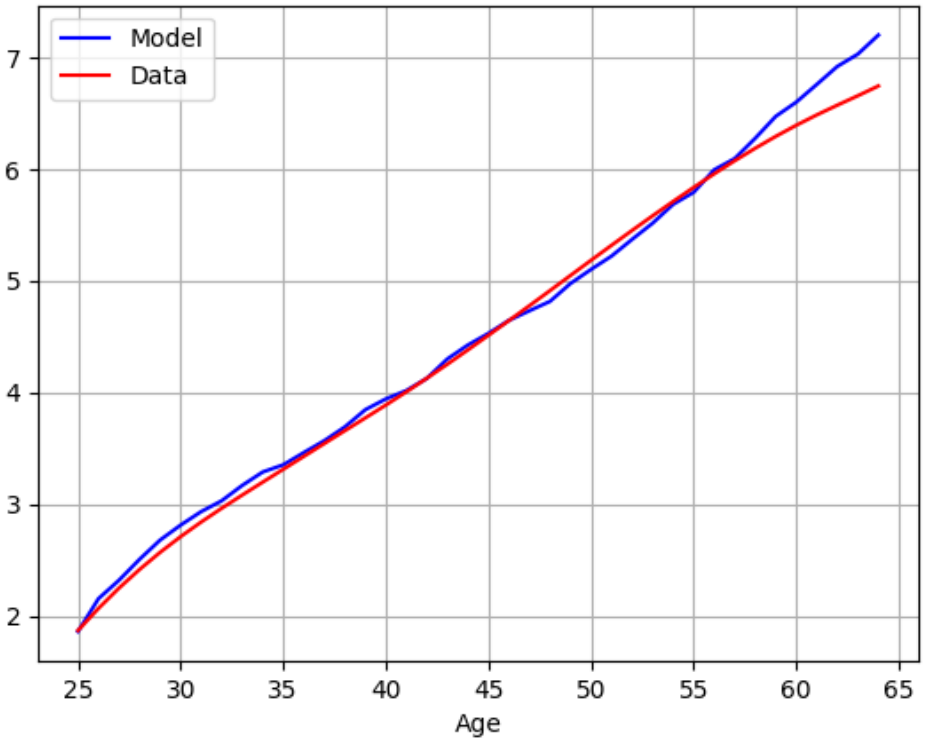
By assumption, all individuals start with a wealth-to-income ratio equal to 1.86. That is why the model and the data exactly overlap at the beginning. The estimated risk aversion and

Table F5: Model Fit and Parameter Estimates

Moments	**Model**	**Data**
Consumption Difference	-7.49809	-7.91000
Mean Wealth Age 25-29	2.23035	2.42337
Mean Wealth Age 30-29	3.03636	3.12760
Mean Wealth Age 35-39	3.57245	3.57651
Mean Wealth Age 40-44	4.12957	3.95796
Mean Wealth Age 45-49	4.71319	4.73432
Mean Wealth Age 50-54	5.34642	5.60820
Mean Wealth Age 55-59	6.07411	6.11601
Mean Wealth Age 60-64	6.86282	6.50887

Parameters	**Values**	**Std. Errors**
Risk Aversion γ	1.48347	0.26551
Discount Factor β	0.99227	0.00404
Separation rate s_{COL}	0.04659	0.13980
Separation rate s_{JA}	0.12644	0.08945
Separation rate f	0.42980	0.28639

Figure F3: Model and Data wealth profiles



discount factors are 1.48 and 0.99, respectively. Figure F3 shows that the life cycle profile fits very well: despite its simplicity, the model matches the wealth profile in the working period remarkably well. The estimate is relatively off target for ages above 55. This is probably because, as people enter their late 50s, many choose to retire, so the remaining workers are highly selected. Given that retirement is exogenous in the model, it fails to replicate the wealth profile of these late workers.

Table F6: Regression with simulated data

Dependent Variable:	Log of Consumption × 100	
	(1)	(2)
Jobs Act	-7.498***	-3.817***
	(0.132)	(0.132)
Age	2.323***	2.239***
	(0.006)	(0.006)
PDV (in €1000)		0.025***
		(0.000)
Observations	168957	168957

Table F6 replicates the empirical analysis in a simulation of the life cycle of 5000 individuals. The first column shows that regressing log consumption on the Jobs Act dummy and age yields the target coefficient. However, when we add the PDV of income as a control, the effect falls to roughly 4%, about half the reduced effect.

F5 Details on the Estimation Algorithm and Inference

Define Θ as the set of estimating parameters, m the vector of target moments, and $g(\Theta, \psi)$ the model counterpart. I estimate Θ by minimizing the following loss function:

$$\hat{\Theta} = \arg\min_{\Theta} \Big[g(\Theta, \psi) - m\Big]' W \Big[g(\Theta, \psi) - m\Big]$$

where W is the inverse of the variance-covariance matrix of the empirical moments, obtained by bootstrapping 1000 samples of the baseline microdata used to compute the moments.

The standard errors of $\hat{\Theta}$ are computed using the Jacobian of the vector of moments G, defined as the derivatives of how model moments change if the parameters change:

$$G(\hat{\Theta}) = \partial \widehat{g(\Theta)} / \partial \Theta$$

The standard errors are then:

$$se(\hat{\Theta}) = \sqrt{\left[G'(\hat{\Theta})WG(\hat{\Theta})\right]^{-1}}$$

Appendix G: Data Collection in detail

G1 Participation, Labor and Unemployment Survey (PLUS)

The *Participation, Labor and Unemployment Survey* (PLUS) is a representative cross-section of the Italian population aged 18 to 75. You can request access to the data on the INAPP website: https://www.inapp.gov.it/.

It includes demographics, information on the number of employees in the firms workers work for, and information on whether they were hired before or after March 7, 2015. In the main analysis, I include only private-sector workers employed in large firms in 2018 with open-ended contracts. Below is a list of variables I use in the analysis:

- **Food Consumption to Family Income Ratio:** The percentage of family income spent on food consumption at home.
- **Family Income:** Average monthly net family income.
- **Food Consumption:** Food Consumption to Family Income Ratio × Family Income (for the imputation)
- **Tenure:** How many years the respondent has been working for the current employer (in bins):
  1. less than 5 years
  2. 5 - 10
  3. 10 - 14
  4. 15 - 19
  5. 20 - 29
  6. 30 - 39
  7. 40 - 49
  8. more than 50
- **Age** (in bins):
  1. 18 - 24
  2. 25 - 29
  3. 30 - 39

4. 40 - 49
5. 50 - 64

G2 Household Budget Survey (HBS)

The *Household Budget Survey* (HBS) is a yearly repeated cross-section that reports information on household spending composition. It details durable and nondurable consumption, as well as demographic information on household composition. It's publicly available at the ISTAT website https://www.istat.it/. HBS is available from 2013 to 2024. I use the 2019 wave to impute consumption into PLUS following Blundell et al. (2008) and Patterson (2023).

G3 Labor Force Survey (LFS)

The textit Labor Force Survey (LFS) is a quarterly survey that provides detailed information on workers' demographics, employment status, tenure, and whether they work for a large or small firm (above or below 15 employees). There are two versions of LFS:

You can easily download the LFS data from the ISTAT website: https://www.istat.it/.

I use all quarterly waves from 2008 to 2019, and I restrict the sample to include working-age individuals hired in the private sector with an open-ended contract. My sample includes almost 1.3 million pooled individual-quarter observations. I use the following variables:

- **Large:** Dummy equal to 1 if the individual works in a firm with at least 15 employees
- **Jobs Act:** Dummy equal to 1 if the individual was starting from March 7, 2015
- **Tenure:** The time difference (in years) between the current year and the year the respondent started her current occupation

Table G1: Summary Statistics

	Mean	Sd	Min	P50	Max
Jobs Act	0.07	0.25	0.00	0	1
Large Firm	0.59	0.49	0.00	1	1
Tenure	14.31	10.63	0.00	12	50

G4 Survey of Household Income and Wealth (SHIW)

The *Survey of Household Income and Wealth* (SHIW) is a biannual survey conducted by the Bank of Italy. Data and documentation are available at https://www.bancaditalia.it/.

I use the dataset to calibrate the model to match the wealth-to-income ratio. The SHIW contains detailed information on demographics, income, wealth, and consumption. The SHIW dataset has two versions: a historical dataset and yearly waves. I use a combination of both, focusing on the 2016 cross-section, which surveys around 8k households and 16k individuals. I restrict the sample to private-sector employees with open-ended contracts. This yields a sample size of around 2k individuals. The Table below shows the descriptive statistics of my sample. The variables I use are the following:

- **Liquid Wealth**: Bank Accounts + Government Bonds + Corporate Bonds
- **Illiquid Wealth**: Stocks + Managed Funds + Foreign Assets + Real Estate
- **Income**: Employee Income + Self-Employed Income
- **Liabilities**: Loans + Mortgages + Other Debts
- **Total Wealth to Income**: Liquid + Illiquid Wealth - Liabilities / Income
- **Marginal Propensity to Consume**: the answer to the following question (used in Jappelli and Pistaferri 2014, 2020 and Auclert et al. 2024):

> *"Imagine you unexpectedly receive a reimbursement equal to the amount your household earns in a month. How much of it would you save, and how much would you spend? Please give the percentage you would save and the percentage you would spend."*

Table G2: Summary Statistics of 2016 SHIW

	Mean	Sd	P50	Min	Max
Consumption	25.52	14.36	22.31	43.30	174.70
Net Wealth	171.94	288.89	121.93	391.00	6501.48
Liquid Wealth	13.29	53.70	4.19	27.00	1945.61
Government Bonds	0.04	0.99	0.00	0.00	32.25
Bank Accounts	11.20	41.18	4.00	21.18	1323.94
Corporate Bonds	2.06	19.41	0.00	0.00	621.68
Illiquid Wealth	161.96	243.54	122.50	357.00	5162.03
Real Estate	159.13	218.49	122.50	355.00	3880.00
Stocks	1.60	33.04	0.00	0.00	1381.40
Managed Wealth	1.00	27.84	0.00	0.00	1235.63
Foreign Wealth	0.22	4.45	0.00	0.00	160.00
Other Assets	0.02	0.47	0.00	0.00	20.00
Liabilities	12.29	37.33	0.00	45.00	476.00
Familiy Labour Earnings	27.01	16.40	21.69	47.00	183.00
Net Wealth to Earnings	6.44	10.28	4.44	14.62	208.66
Illiquid Wealth to Earnings	6.12	9.13	4.40	13.70	135.72
Liquid Wealth to Earnings	0.48	2.17	0.17	0.92	66.18
Personal Labour Earnings	18.91	9.48	18.00	27.49	130.00
Marginal Propensity to Consume	0.48	0.34	0.50	1.00	1.00
Wealthy HtM	0.12	0.33	0.00	1.00	1.00
HtM	0.23	0.42	0.00	1.00	1.00

Notes: The monetary variables are expressed in 2016 €1000

Figure G1: Life Cycle Profile of Consumption, MPC, and Wealth

